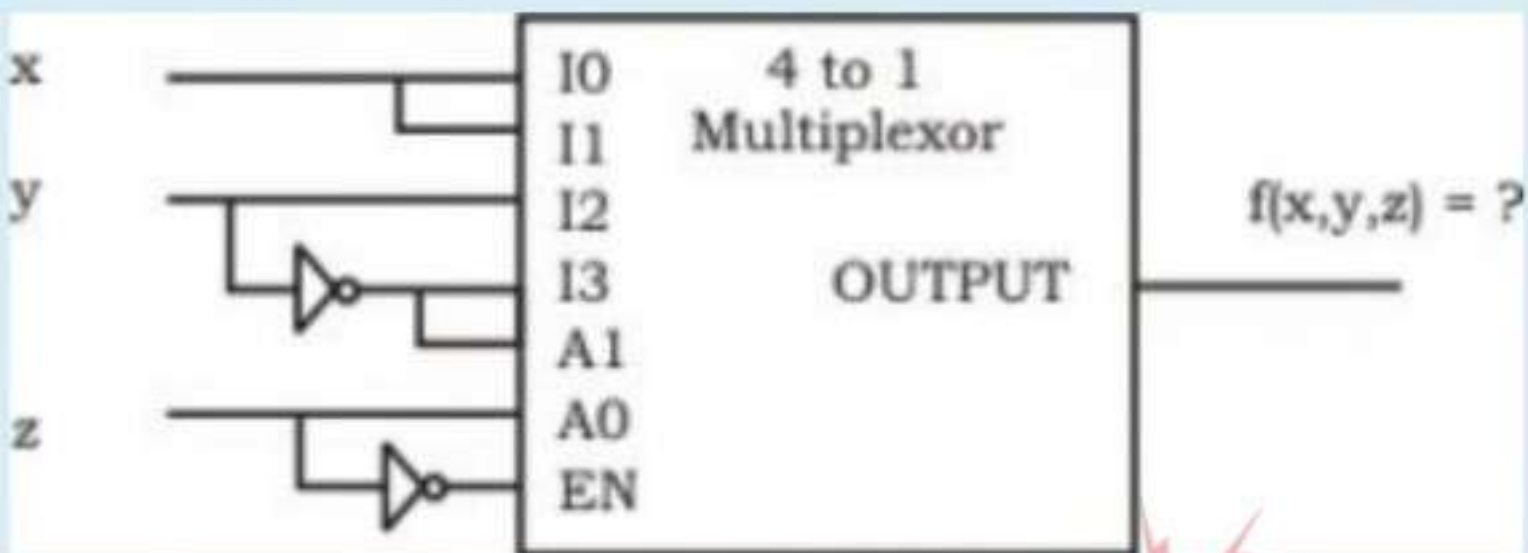




Spar

POLYTECHNIC

=F

اختر أحد الخيارات

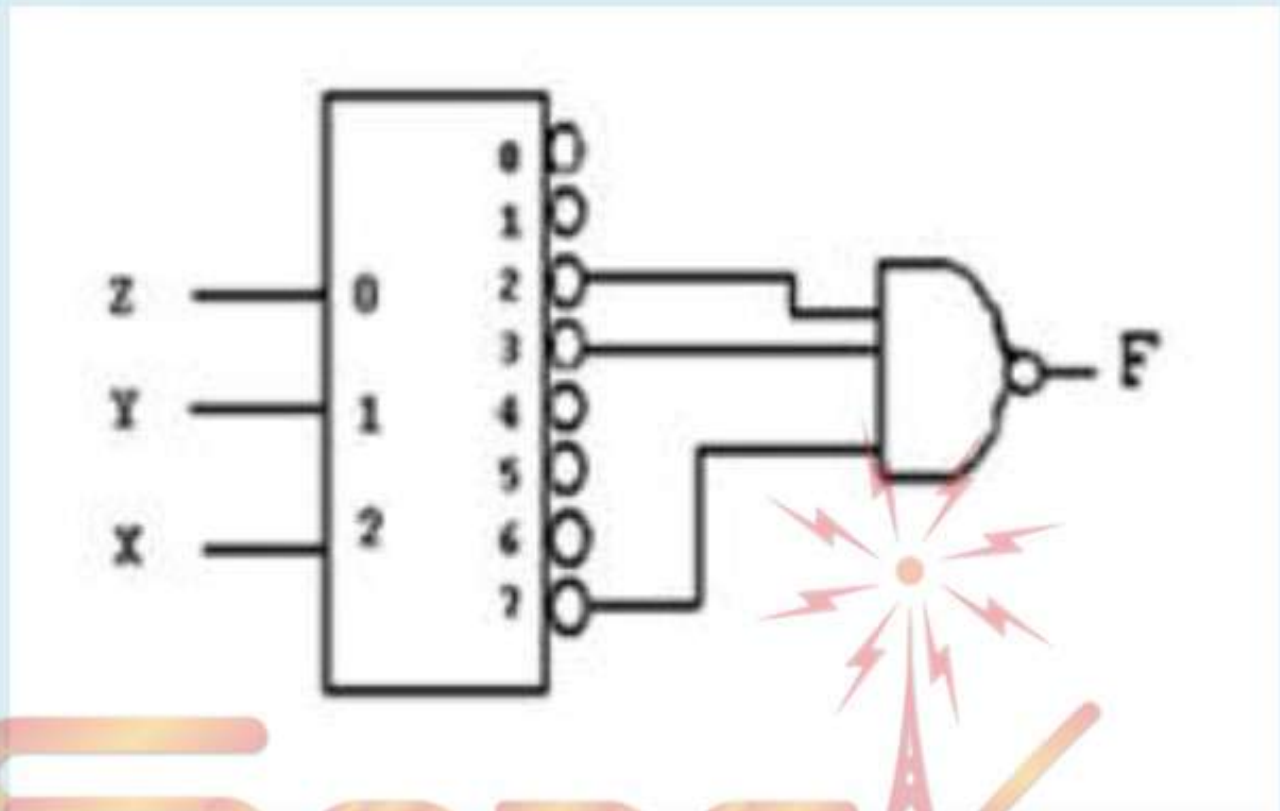
None of all ☐

'XYZ ☐

X+Z ☐

XY+Z ☐

The Function $F(x,y,z)$ represented by the following decoder (decoder outputs are active low)



SPAR
POLYTECHNIC

اختر أحد الخيارات

a. $\Sigma (0, 1, 4, 5, 6)$ ☐

b. $\pi (0, 1, 4, 5, 6)$ ☐

c. $\Sigma (2, 3, 7)$ ☐

d. $\pi (2, 3, 7)$ ☐

When Design a counter that counts in the sequence XYZ : 001 --> 011--> 010--> 110-> 111--> 101 and repeat , using SR flip flops :After you COMPLETE the table below

Answer the following questions



The boolean function for **SX** is .1



The boolean function for **RX** is .2



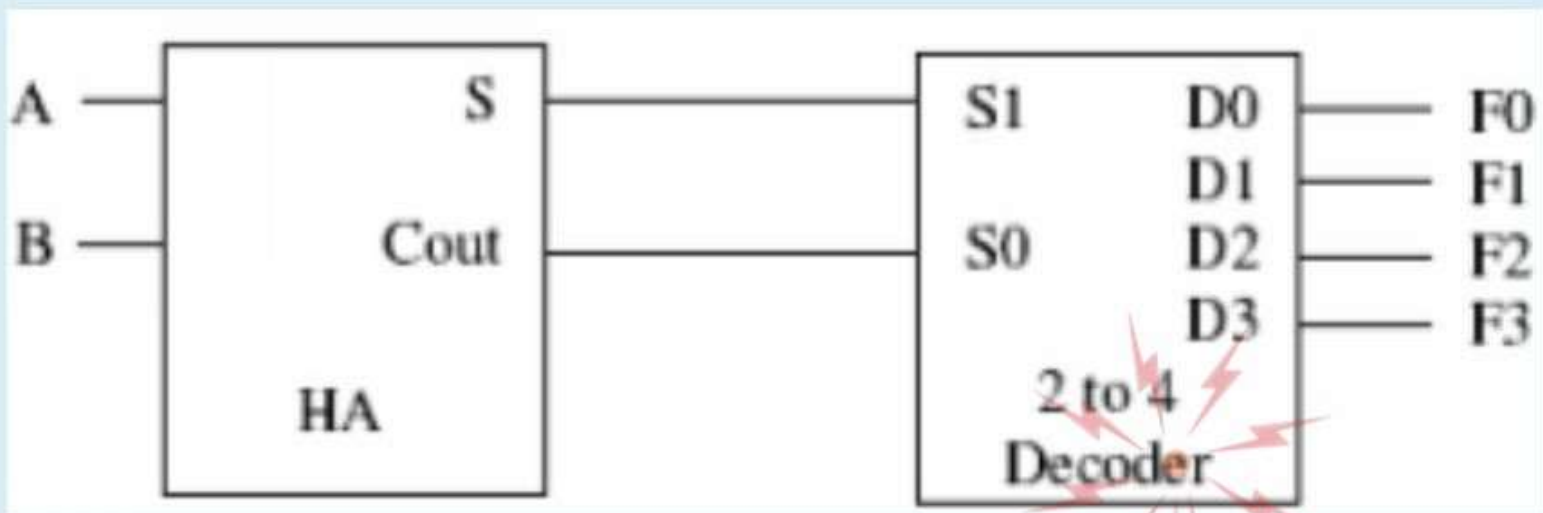
The boolean function for **SY** is .3

How many **SR** Flip Flop you must use to .4



?design the circuit

The figure below shows a Half-Adder connected to a 2 to 4 decoder



if the output (F0 F1 F2 F3) is (0 1 0 0), what is the inputs (A B)

Spar
POLYTECHNIC

اختر أحد الخيارات

a. 1 1 ☐

b. 0 0 ☐

c. 0 1 ☐

d. 1 0 ☐

Design a Grade-to-7 Segment display decoding circuit that will perform the following Grade-to-Symbol conversion

Decimal Input	7-Segment Output	
0	F	
1	E	
2	C	
3	B	

?which the following truth table is correct

اختر أحد الخيارات
☐ a

Inputs		7-Segment							Display
A1	A0	a	b	c	d	e	f	g	
0	0	1	1	1	1	1	1	0	F
0	1	0	0	0	0	1	1	0	E
1	0	1	1	0	1	1	0	1	C
1	1	1	1	1	1	0	0	1	B

☐ b

Inputs		7-Segment							Display
A1	A0	a	b	c	d	e	f	g	
0	0	1	0	0	0	1	1	1	F
0	1	1	0	0	1	1	1	1	E
1	0	1	0	0	1	1	1	0	C
1	1	1	1	1	1	1	1	1	B

☐ c

Inputs		7-Segment							Display
A1	A0	a	b	c	d	e	f	g	
0	0	0	0	0	0	0	0	1	F
0	1	1	1	1	1	0	0	1	E
1	0	0	0	1	0	0	1	0	C
1	1	0	0	0	0	1	1	0	B

An RS flip-flop is the basic type of



sequential logic circuits

?What is a sequential circuit

اخترأحد الخيارات

a. a circuit composed of a sequence of elementary logic gates, e.g. AND gates ☐

b. a circuit containing memory elements, in which the outputs not only depend on the input but also on the state of the memory element ☐

c. a circuit in which the outputs depend only on the inputs at any moment ☐

d. a loop circuit, always repeating the same cycle ☐

What is the largest number of data inputs
which a data selector with three control inputs
? can have



اختر أحد الخيارات

a. 2 ☐

b. 8 ☐

c. 4 ☐

d. 16 ☐

In a comparator, if we get input as $A > B$ then _____ the output will be

اختراع الخيارات



1 ☐

A ☒

0 ☐

B ☐

In a **comparator**, if we get input as $A > B$ then _____ the output will be

اختر أحد الخيارات

1 ☐

A ☐

0 ☐

B ☐



What additional feature have a sequential circuit, compared to a combinational circuit

اختراع الخيارات

a. a memory state ☐

b. AND or OR gates ☐

c. clock pulse generators ☐

d. a power transistor ☐

The number of full and half-adders required to add 8-bit numbers is

اختر أحد الخيارات

a. 1 half-adder, 7 full-adders ☒

b. 4 half-adders, 4 full-adders ☐

c. half-adders, 6 full-adders ☐

d. 7 half-adders, 1 full-adders ☐

DIGITAL LOGIC DESIGN

A circuit that compares two numbers and determine their magnitude is called _____

Select one:

- ☒ Magnitude comparator
- ☐ Height comparator
- ☐ Size comparator
- ☐ Comparator

Spark

DIGITAL LOGIC DESIGN LAB-8

Don't care conditions can be used for simplifying Boolean expressions in _____

Select one:

- ☐ Latches
- ☐ Terms
- ☐ K-maps
- ☐ Registers

Spark

DIGITAL LOGIC DESIGN LAB-8

What logic gate that generates an output of logic zero (LOW) only when all its inputs are logic one (HIGH)?

Select one:

- ☐ OR gate
- ☐ NOR gate
- ☐ AND gate
- ☐ NAND gate

Spark

13-DIGITAL LOGIC DESIGN LAB-8

Question 14

Not yet answered

Started out of 1.00

Flag question

How many INVERTERS circuits are contained in a 7404 IC?

Select one:

0

6

2

4

Spark

DIGITAL LOGIC DESIGN LAB-0

There are _____ Minterms for 3 variables (a, b, c).

Select one:

☐ 1

☐ 0

☒ 8

Spark

Question 6

Not yet answered

Marked out of 1.00

Flag question

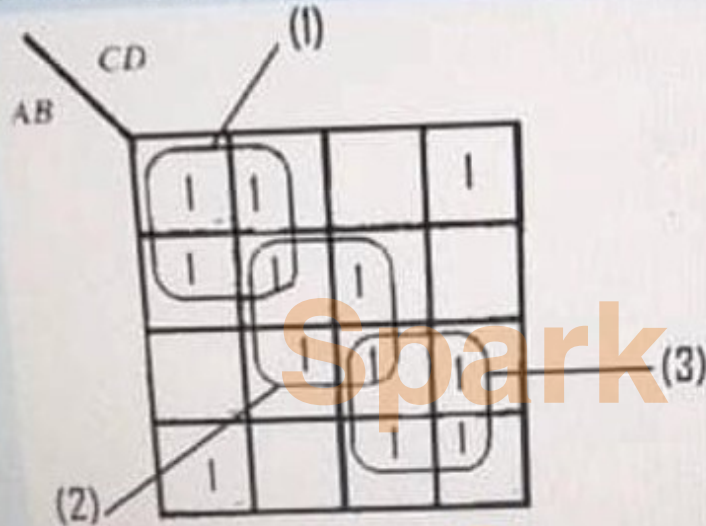
Which statement below best describes a Karnaugh map?

Select one:

- ☒ It is simply a rearranged truth table
- ☐ The Karnaugh map eliminates the need for using NAND and NOR gates
- ☐ A Karnaugh map can be used to replace Boolean rules
- ☐ Variable complements can be eliminated by using Karnaugh maps

DIGITAL LOGIC DESIGN LAB-8

Depending on the K-map Loop 3 gives



AC

Select one:

BO

B'D

AC

A'C'

Q.8
answered
out of 1.00
g question

In which operation carry is obtained?

Select one:

- ☐ Multiplication
- ☒ Both addition and subtraction
- ☐ Addition
- ☐ Subtraction

Spark

DIGITAL LOGIC DESIGN

DeMorgan's theorem states that _____

Select one

☐ $A' + B' = A'B'$

☐ $(A + B)' = A' + B'$

☒ $(AB)' = A' + B'$

☐ $(AB) = A + B$

Spark

DIGITAL LOGIC DESIGN

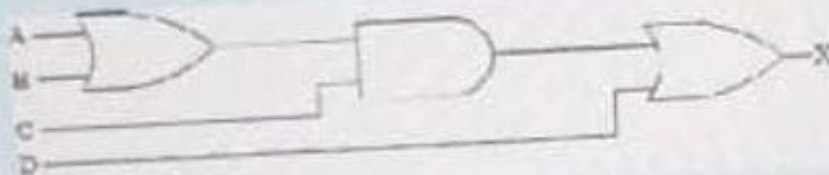
Question 3

Unanswered

0 out of 1.00

Flag Question

Solve the circuit for X.



Select one

☒ $((A + B)C) + D$

☐ $A + BC + D$

☐ $D(A + B + C)$

☐ $(AC + BC)D$

Question 9

Incorrect

Mark 0.00 out of
1.00

Flag question

All numbers from 0 to 15 are represented in a BCD-7-segment decoder driver.

Select one:

- ☒ a. the question wrong ✖
- ☐ b. no answer
- ☐ c. True
- ☐ d. False

The correct answer is: False

Question 10

Correct

Mark 1.00 out of
1.00

Flag question

Which logic gates can be used to make a flip-flop?

Select one:

- ☐ a. OR
- ☒ b. 1 NAND ✔
- ☐ c. AND
- ☐ d. EXCLUSIVE OR

The correct answer is: 1 NAND

Question 11

Not answered

Marked out of 1.00

Flag question

What is a sequential circuit?

Select one:

- a. a circuit composed of a sequence of elementary logic gates, e.g. AND gates
- b. a circuit in which the outputs depend only on the inputs at any moment
- c. a circuit containing memory elements, in which the outputs not only depend on the input but also on the s

Question 19

Not answered

Marked out of 1.00

Flag question

a 7-segment display shows:

Select one:

- ☐ a. only the numbers from 0 to 9
- ☐ b. all alphanumeric characters
- ☐ c. all the letters of the alphabet
- ☐ d. all numbers from 0 to 9, the letters A, b, c, d, E, F and some other symbols

The correct answer is: all numbers from 0 to 9, the letters A, b, c, d, E, F and some other symbols

Spark

Question 20

Not answered

Marked out of 1.00

Flag question

What additional feature have a sequential circuit, compared to a combinational circuit?

Select one:

- ☐ a. clock pulse generators
- ☐ b. AND or OR gates
- ☐ c. a power transistor
- ☐ d. a memory state

The correct answer is: a memory state

Question 21

Not answered

Marked out of 1.00

Flag question

What is the function of a binary-decimal coder?

Select one:

- ☐ a. to input data in binary way into a Analog system
- ☐ b. to input data in binary way into a digital system
- ☐ c. to count
- ☐ d. to convert a binary number into decimal one

The correct answer is: to convert a binary number into decimal one

Question 22

Not answered

Marked out of 1.00

Flag question

How is numerical data displayed normally, in digital systems?

Select one:

- ☐ a. with a 7-segment display
- ☐ b. with a counter
- ☐ c. with a multimeter
- ☐ d. with a speaker

The correct answer is: with a 7-segment display

Question 22

Not answered

Marked out of 1.00

Flag question

How is numerical data displayed normally, in digital systems?

Select one:

- ☐ a. with a 7-segment display
- ☐ b. with a counter
- ☐ c. with a multimeter
- ☐ d. with a speaker

The correct answer is: with a 7-segment display

Question 23

Not answered

Marked out of 1.00

Flag question

How many inputs and outputs has a BCD-decimal coder SN74LS42 got?

Select one:

- ☐ a. 9 inputs and 4 outputs
- ☐ b. 4 inputs and 10 outputs
- ☐ c. 10 inputs and 4 outputs
- ☐ d. 4 inputs and 9 outputs

The correct answer is: 4 inputs and 10 outputs

11

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estion

What is a sequential circuit?

Select one:

- ☐ a. a circuit composed of a sequence of elementary logic gates, e.g. AND gates
- ☐ b. a circuit in which the outputs depend only on the inputs at any moment
- ☐ c. a circuit containing memory elements, in which the outputs not only depend on the input but also on the state of the memory element
- ☐ d. a loop circuit, always repeating the same cycle

The correct answer is: a circuit containing memory elements, in which the outputs not only depend on the input but also on the state of the element.

12

ered

it of 2.00

uestion

How many flip flop we have?

Answer:



The correct answer is: 4

13

ered

ut of 1.00

uestion

Which kind of code must be supplied to the inputs of a BCD-to-decimal coder?

Select one:

- a. Gray
- b. pure binary
- c. BCD

Question 13

Not answered

Marked out of 1.00

Flag question

Which kind of code must be supplied to the inputs of a BCD-to-decimal coder?

Select one:

- ☐ a. Gray
- ☐ b. pure binary
- ☐ c. BCD
- ☐ d. excess 3

The correct answer is: BCD

Question 14

Not answered

Marked out of 1.00

Flag question

Which is the better type of drive for an led display?

Select one:

- ☐ a. Current
- ☐ b. Micro
- ☐ c. Voltage
- ☐ d. R

The correct answer is: Current

Question 15

Not answered

Marked out of 1.00

Flag question

A binary-decimal coder is composed of:

Select one:

- ☐ a. logic gates only
- ☐ b. logic gates and flip-flops
- ☐ c. flip-flops only

Question 15

Not answered

Marked out of 1.00

Flag question

A binary-decimal coder is composed of:

Select one:

- ☐ a. logic gates only
- ☐ b. logic gates and flip-flops
- ☐ c. flip-flops only
- ☐ d. none of them

The correct answer is: logic gates only

Question 16

Correct

Mark 1.00 out of 1.00

Flag question

Which kind of display is it better to use for the display of the alphanumeric characters?

Select one:

- ☒ a. 14-16-18-segment display ✓
- ☐ b. nixie display
- ☐ c. 7-segment display
- ☐ d. dot-matrix display (e.g. 6 x 8)

The correct answer is: 14-16-18-segment display

Question 17

Not answered

Marked out of 1.00

Flag question

What do you mean, when you say that a flip-flop is an indeterminate state?

Select one:

- ☐ a. that after switching flip-flop, the memory state is undefined, and can be $Q=0$ and $Q=1$ or $Q=1$ and $Q=0$

Question 17

Not answered

Marked out of 1.00

Flag question

What do you mean, when you say that a flip-flop is an indeterminate state?

Select one:

- ☐ a. that after switching flip-flop, the memory state is undefined, and can be $Q=0$ and $Q'=1$ or $Q=1$ and $Q'=0$
- ☐ b. that the outputs are not at level 0, nor 1
- ☐ c. that the outputs are not fired
- ☐ d. that the flip-flop tends to oscillate indefinitely

The correct answer is: that after switching flip-flop, the memory state is undefined, and can be $Q=0$ and $Q'=1$ or

Question 18

Not answered

Marked out of 1.00

Flag question

In a 7-segment display decoder driver using negative logic (common anode display), a low output level represents

Select one:

- ☐ a. False
- ☐ b. True
- ☐ c. the question wrong
- ☐ d. No answer

The correct answer is: True

Question 19

Not answered

Marked out of 1.00

Flag question

a 7-segment display shows:

Select one:

- ☐ a. only the numbers from 0 to 9
- ☐ b. all alphanumeric characters
- ☐ c. all the letters of the alphabet
- ☐ d. all the numbers from 0 to 9, the letters A to Z, and the special characters

Question 3

Correct

Mark 1.00 out of 1.00

Flag question

An RS flip-flop is the basic type of sequential logic circuits. ✓

Your answer is correct.

The correct answer is:

An RS flip-flop is the basic type of [sequential logic circuits.]

Question 4

Correct

Mark 1.00 out of 1.00

Flag question

When happens a state change in a clocked sequential circuit?

Select one:

- ☐ a. when the clock signal is high
- ☐ b. at any moment: it depends on the inputs
- ☒ c. on the rising or falling edges of the clock ✓
- ☐ d. during the period "T" of the clock

The correct answer is: on the rising or falling edges of the clock

Question 5

Not answered

Marked out of 1.00

Flag question

In a coder with priority (Priority Encoder):

Select one:

- ☐ a. the input with heavier (or lighter) weight is considered active
- ☐ b. the output is correct only if the inputs are active one at a time
- ☐ c. only the input activated first is considered active

Question 5

Not answered

Marked out of 1.00

Flag question

In a coder with priority (Priority Encoder):

Select one:

- ☐ a. the input with heavier (or lighter) weight is considered active
- ☐ b. the output is correct only if the inputs are active one at a time
- ☐ c. only the input activated first is considered active
- ☐ d. when the inputs are active, the outputs are not

The correct answer is: the input with heavier (or lighter) weight is considered active

Question 6

Correct

Mark 2.00 out of 2.00

Flag question

How many represents a number of output lines for a decoder with 4 input lines

Answer: 16

The correct answer is: 16

Question 7

Incorrect

Mark 0.00 out of 1.00

Flag question

What is the function of a decimal-binary encoder?

Select one:

- ☒ a. to convert from binary to decimal code X
- ☐ b. to convert a BCD number into an hexadecimal one
- ☐ c. to convert an analog signal into a digital one
- ☐ d. to convert from decimal to binary code

Question 7

Incorrect

Mark 0.00 out of

1.00

Flag question

What is the function of a decimal-binary encoder?

Select one:

- ☒ a. to convert from binary to decimal code **X**
- ☐ b. to convert a BCD number into an hexadecimal one
- ☐ c. to convert an analog signal into a digital one
- ☐ d. to convert from decimal to binary code

The correct answer is: to convert from decimal to binary code

Question 8

Not answered

Marked out of 2.00

Flag question

With common cathode, driven with

Answer:

The correct answer is: positive logic

Question 9

Incorrect

Mark 0.00 out of

1.00

All numbers from 0 to 15 are represented in a BCD-7-segment decoder driver.

Select one:

- ☒ a. the question wrong **X**
- ☐ b. no answer

on 1

answered

1 out of 1.00

g question

How is numerical data displayed normally, in digital systems?

Select one:

- ☐ a. with a 7-segment display
- ☐ b. with a multimeter
- ☐ c. with a speaker
- ☐ d. with a counter

tion 2

et answered

ed out of 1.00

lag question

What is the function of a decimal-binary encoder?

Select one:

- ☐ a. to convert an analog signal into a digital one
- ☐ b. to convert a BCD number into an hexadecimal one
- ☐ c. to convert from binary to decimal code
- ☐ d. to convert from decimal to binary code

stion 3

yet answered

ked out of 1.00

the question

In a 7-segment display decoder driver using negative logic (common anode display), the ON state.

Select one:

AL LOGIC DESI

Which of the following expressions is in the product-of-sums form?

Select one:

- ☐ $AB(CD)$
- ☐ $AB + CD$
- ☐ $(AB)(CD)$
- ☒ $(A + B)(C + D)$

Spark

DIGITAL LOGIC DESIGN LAB

10

answered

1 out of 1.00

question

If A and B are the inputs of a half adder, the carry is given by _____

Select one

☐ A EX-NOR B

☐ A OR B

☐ A XOR B

☒ A AND B

Spark

AL LOGIC

How many NAND circuits are contained in a 7400 NAND IC?

Select one:

☐ 8

☐ 2

☒ 4

Spark

DIGITAL LOGIC DESIGN

17

answered

out of 1.00

question

How many AND circuits are contained in a 7408 AND IC?

Select one:

☒ 2

☐ 4

☐ 8

Spark

Question 17

Incorrect

Mark 0.00 out of
1.50

Flag question

A demultiplexer is used to

Select one:

- ☐ a. Perform serial to parallel conversion
- ☒ b. Route the data from single input to one of many outputs ✗
- ☐ c. Both route the data and perform serial to parallel
- ☐ d. Select data from several inputs and route it to single output

The correct answer is: Both route the data and perform serial to parallel

Question 19

Incorrect

Mark 0.00 out of
1.50

Flag question

If the input J is connected through K input of J-K, then flip-flop will behave as a

Select one:

- ☐ a. D type flip-flop
- ☐ b. T type flip-flop
- ☐ c. S-R flip-flop
- ☒ d. Master slave JK flip-flop ✗

The correct answer is: T type flip-flop

Question 22

Incorrect

Mark 0.00 out of
1.50

Flag question

Which of the following adders can add two binary numbers at a time ?

Select one:

- ☐ a. Carry-save-adder
- ☐ b. Parallel adder
- ☐ c. Full adder
- ☒ d. Carry-look-ahead adder 

The correct answer is: Full adder

Question 11

Incorrect

Mark 0.00 out of
1.50

Flag question

To implement the Boolean function $F = A \oplus B \oplus C$ we can use a 3-to-8 decoder and connect:

Select one:

- a. 3 different lines from its output to an OR gate
- b. 6 different lines from its output to an OR gate
- ☒ c. 5 different lines from its output to an OR gate ✗
- d. 4 different lines from its output to an OR gate

Question 4

Incorrect

Mark 0.00 out of
1.50

Flag question

A combinational logic circuit which generates a particular binary word or number is

Select one:

- ☒ a. Multiplexer ✗
- ☐ b. Demultiplexer
- ☐ c. Decoder
- ☐ d. Encoder

The correct answer is: Decoder

Question 11

Incorrect

Mark 0.00 out of
1.50

Flag question

To implement the Boolean function $F = A \oplus B \oplus C$ we can use a 3-to-8 decoder and connect:

Select one:

- a. 3 different lines from its output to an OR gate
- b. 6 different lines from its output to an OR gate
- ☒ c. 5 different lines from its output to an OR gate ✗
- d. 4 different lines from its output to an OR gate

How many minimal numbers of basic logic gates (NOT, AND, OR) need to represent the Boolean function ?

Select one:

- ☒ a. Two NOTs, one OR, one AND. ✗
- ☐ b. Two NOT, one OR.
- ☐ c. One NOT, one OR.
- ☐ d. One NOT, one AND.

The correct answer is: One NOT, one OR.

How many inputs and outputs has a BCD-decimal coder SN74LS42 got?

Select one:

- ☐ a. 4 inputs and 9 outputs
- ☒ b. 4 inputs and 10 outputs
- ☐ c. 10 inputs and 4 outputs
- ☐ d. 9 inputs and 4 outputs

What is a sequential circuit?

Select one:

- ☒ a. a circuit containing memory elements, in which the outputs not only depend on the input but also on the state of the memory element.
- ☐ b. a loop circuit, always repeating the same cycle
- ☐ c. a circuit composed of a sequence of elementary logic gates, e.g. AND gates
- ☐ d. a circuit in which the outputs depend only on the inputs at any moment

How many filp flops we have?

Answer:

Spark

4

What do you mean, when you say that a flip-flop is an indeterminate state?

Select one:

- ☐ a. that the outputs are not at level 0, nor 1
- ☒ b. that after switching flip-flop, the memory state is undefined, and can be $Q=0$ and $Q'=1$ or $Q=1$ and $Q'=0$
- ☐ c. that the flip-flop tends to oscillate indefinitely
- ☐ d. that the outputs are not fired

a 7-segment display shows:

Select one:

- ☒ a. all numbers from 0 to 9, the letters A, b, c, d, E, F and some other symbols
- ☐ b. all alphanumeric characters
- ☐ c. only the numbers from 0 to 9
- ☐ d. all the letters of the alphabet

Which is the better type of drive for an led display?

Select one:

- ☐ a. Micro
- ☒ b. Current
- ☐ c. Voltage
- ☐ d. R

Question 7

Not yet answered

Marked out of 1.00

 Flag question

Spark

**AND,OR, NAND and NOR
considered as**

Choose...



There are _____ cells in a 4-variable K-map.

Select one:

- ☐ a. 8
- ☐ b. 18
- ☐ c. 16
- ☐ d. 12



A latch is edge sensitive

Select one:

☐ a. false

☒ b. true

In a 7-segment display decoder driver using positive logic (common cathode display), a low output level represents a segment in the ON state.

Select one:

- ☐ a. TRUE
- ☐ b. FALSE



fill the state table

JKQQ+

1 1 0 ? ? ? ?



Select one:

- ☐ a. 0
- ☐ b. Q
- ☐ c. Q'
- ☐ d. 1

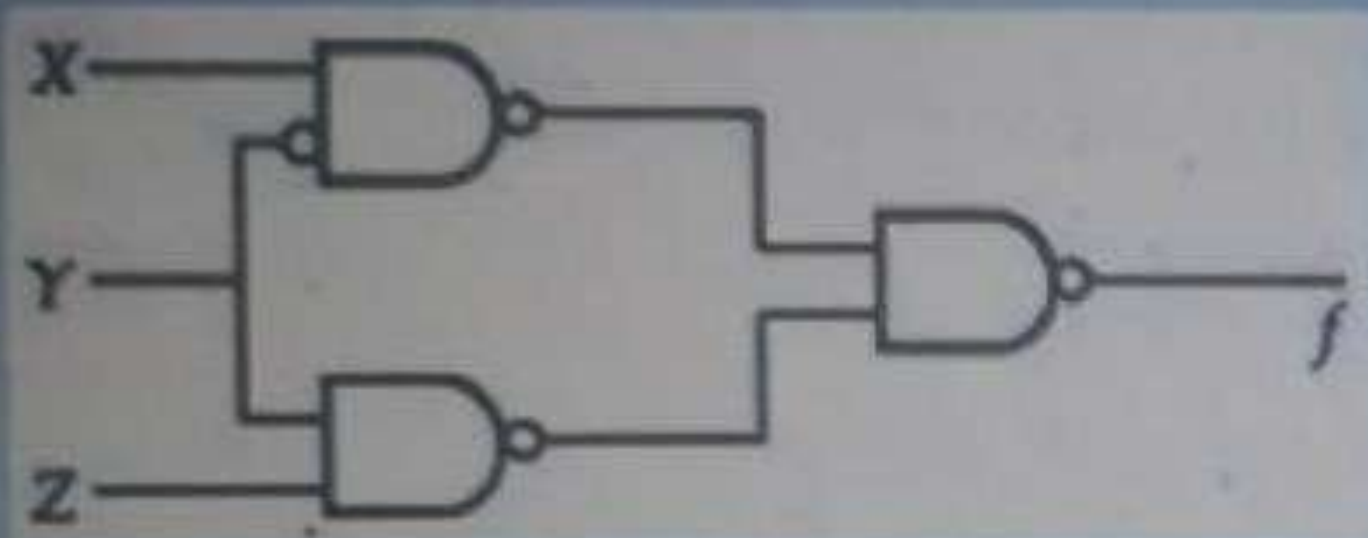
In a 7-segment display decoder driver using positive logic (common cathode display), a low output level represents a segment in the ON state.

Select one:

☐ a. FALSE

☐ b. TRUE





Which one of the following is TRUE?

Select one:

- ☐ a. f is independent of X
- ☐ b. f is independent of Z
- ☐ c. f is independent of Y
- ☐ d. None of X, Y, Z is redundant

In a 7-segment display decoder driver using positive logic (common cathode display), a low output level represents a segment in the ON state.

Select one:

- ☐ a. TRUE
- ☐ b. FALSE



If A, B and C are the inputs of a full adder then the sum is given by _____

Select one:

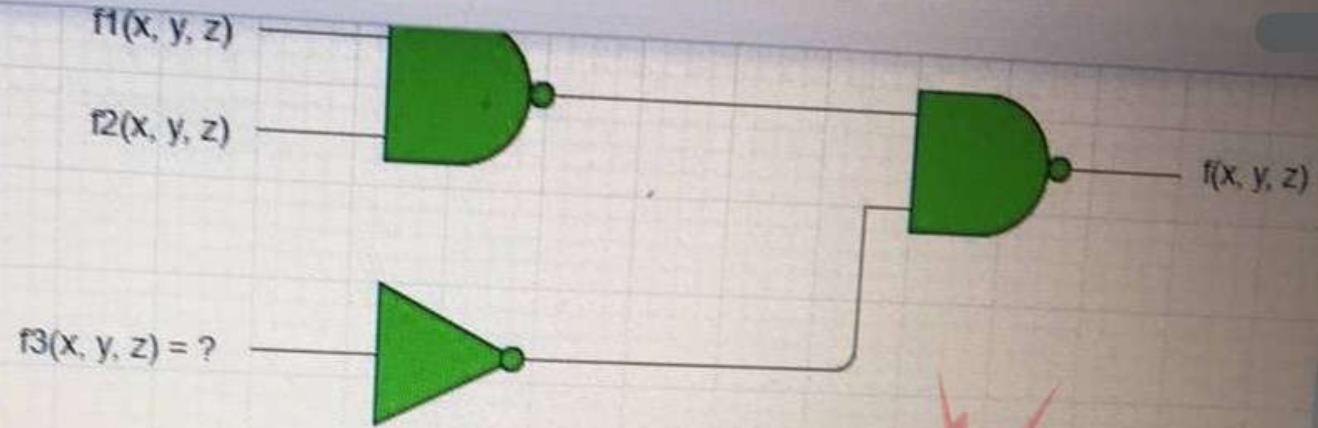
☐ A AND B AND C

☒ A XOR B XOR C

☐ A OR B OR C

☐ A OR B AND C

C



Given that

$f_1(x, y, z) = \Sigma(0, 1, 3, 5),$
 $f_2(x, y, z) = \Sigma(6, 7)$ and
 $f(x, y, z) = \Sigma(1, 4, 5),$

f_3 is :

Select one:

- ☐ a. None of these
- ☐ b. $\Sigma(6, 7)$
- ☐ c. $\Sigma(1, 4, 5)$
- ☐ d. $\Sigma(0, 1, 3, 5)$

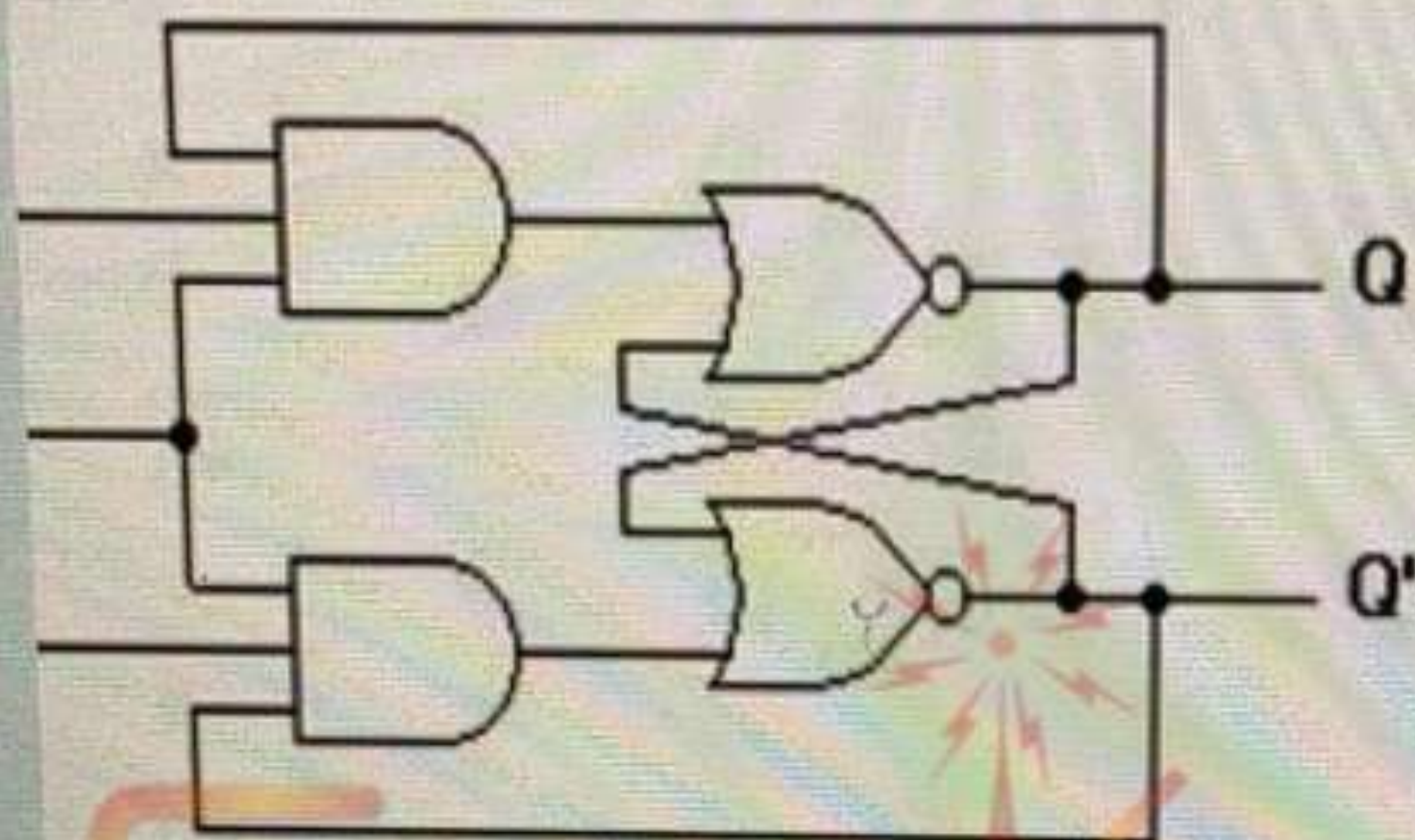
In a 7-segment display decoder driver using positive logic (common cathode display), a low output level represents a segment in the ON state.

Select one:

- ☐ a. TRUE
- ☐ b. FALSE



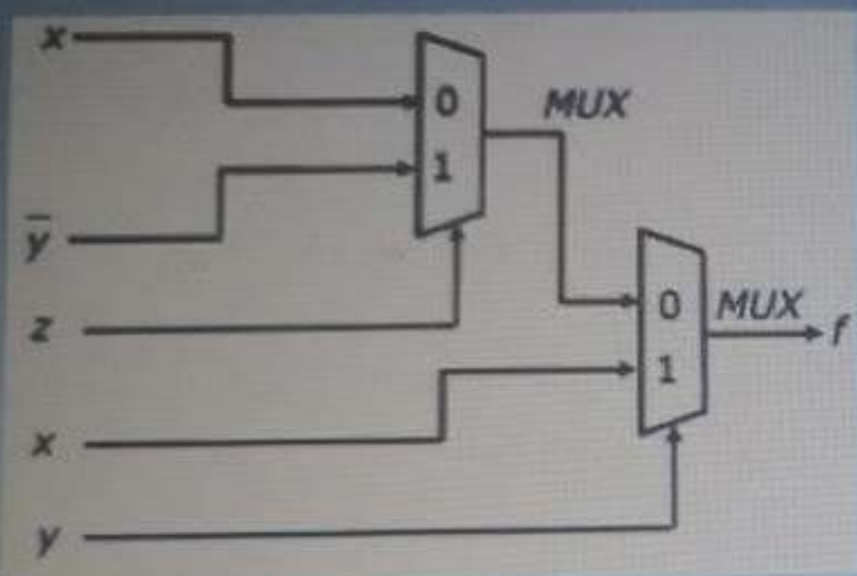
what kind of flip flop is shown



Select one:

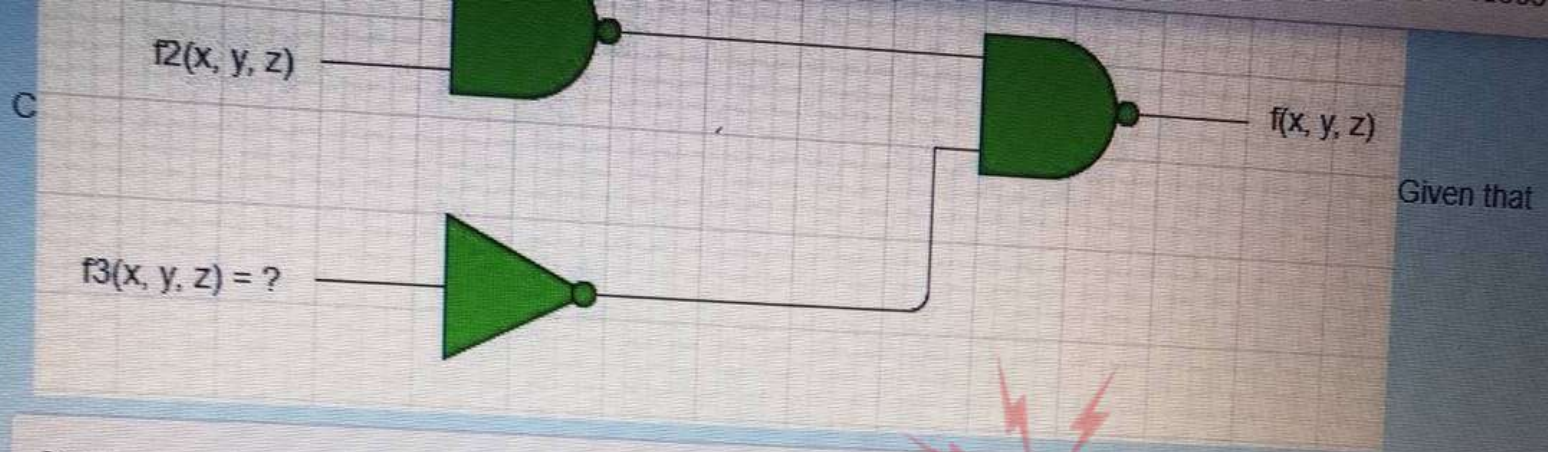
- ☐ a. D-FF
- ☐ b. T-FF
- ☐ c. SR-FF
- ☐ d. JK-FF

Consider the circuit. Which one of the following options correctly represents $f(x, y, z)$?



Select one:

- ☐ a.
 $xz' + xy + (yz)'$
- ☐ b.
 $xz' + xy + y'z$
- ☐ c.
 $xz + xy + (yz)'$
- ☐ d.
 $xz + xy + y'z$



$f_1(dx, y, z) = \Sigma(0, 1, 3, 5),$
 $f_2(dx, y, z) = \Sigma(6, 7)$ and
 $f(dx, y, z) = \Sigma(1, 4, 5),$

f_3 is :

Select one:

☐ a. None of these

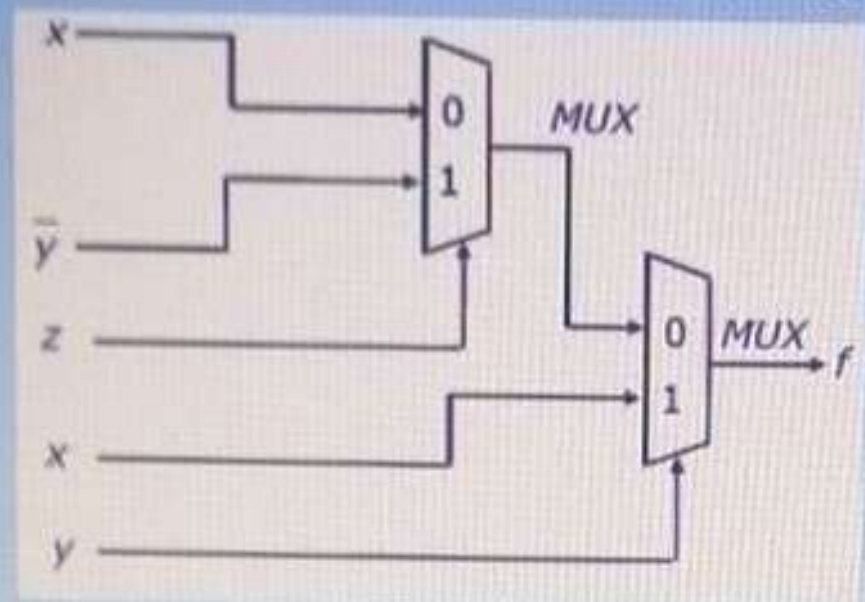
☐ b. $\Sigma(6, 7)$

☐ c. $\Sigma(1, 4, 5)$

☐ d. $\Sigma(0, 1, 3, 5)$

Protected with free version of Watermarkly. Full version doesn't put this mark.

Consider the circuit . Which one of the following options correctly represents $f(x, y, z)$?



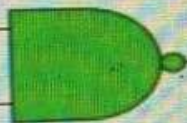
Select one:

- ☐ a.
 $xz' + xy + (yz)'$
- ☐ b.
 $xz + xy + (yz)'$
- ☐ c.
 $xz + xy' + y'z$
- ☐ d.
 $xz' + xy + y'z$

$f_1(x, y, z)$

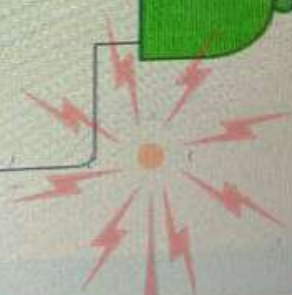
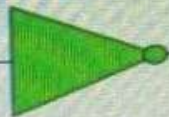
$f_2(x, y, z)$

$f_3(x, y, z) = ?$



$f(x, y, z)$

Given



that

$$f_1(x, y, z) = \sum(0, 1, 3, 5),$$

$$f_2(x, y, z) = \sum(6, 7) \text{ and}$$

$$f(x, y, z) = \sum(1, 4, 5),$$

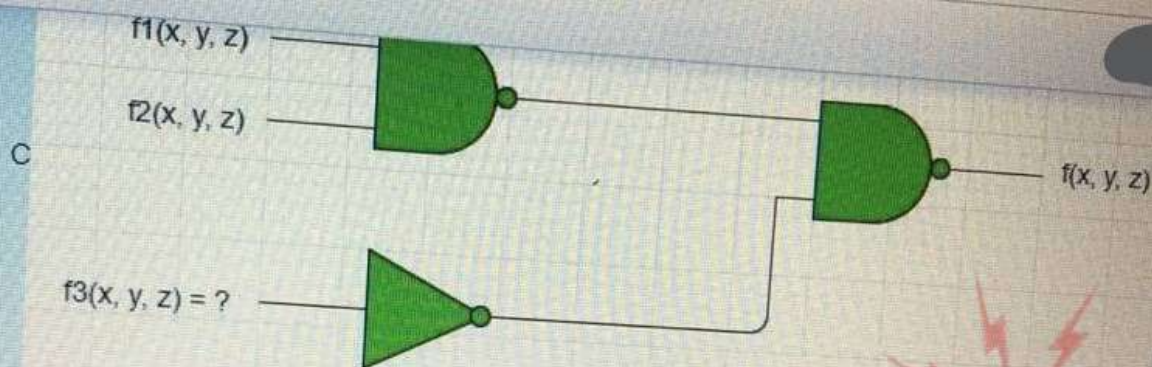
f_3 is

Select one



a

None of these



Given that

$$f1(dx, y, z) = \Sigma(0, 1, 3, 5),$$

$$f2(dx, y, z) = \Sigma(6, 7) \text{ and}$$

$$f(dx, y, z) = \Sigma(1, 4, 5),$$

$f3$ is :

Select one:

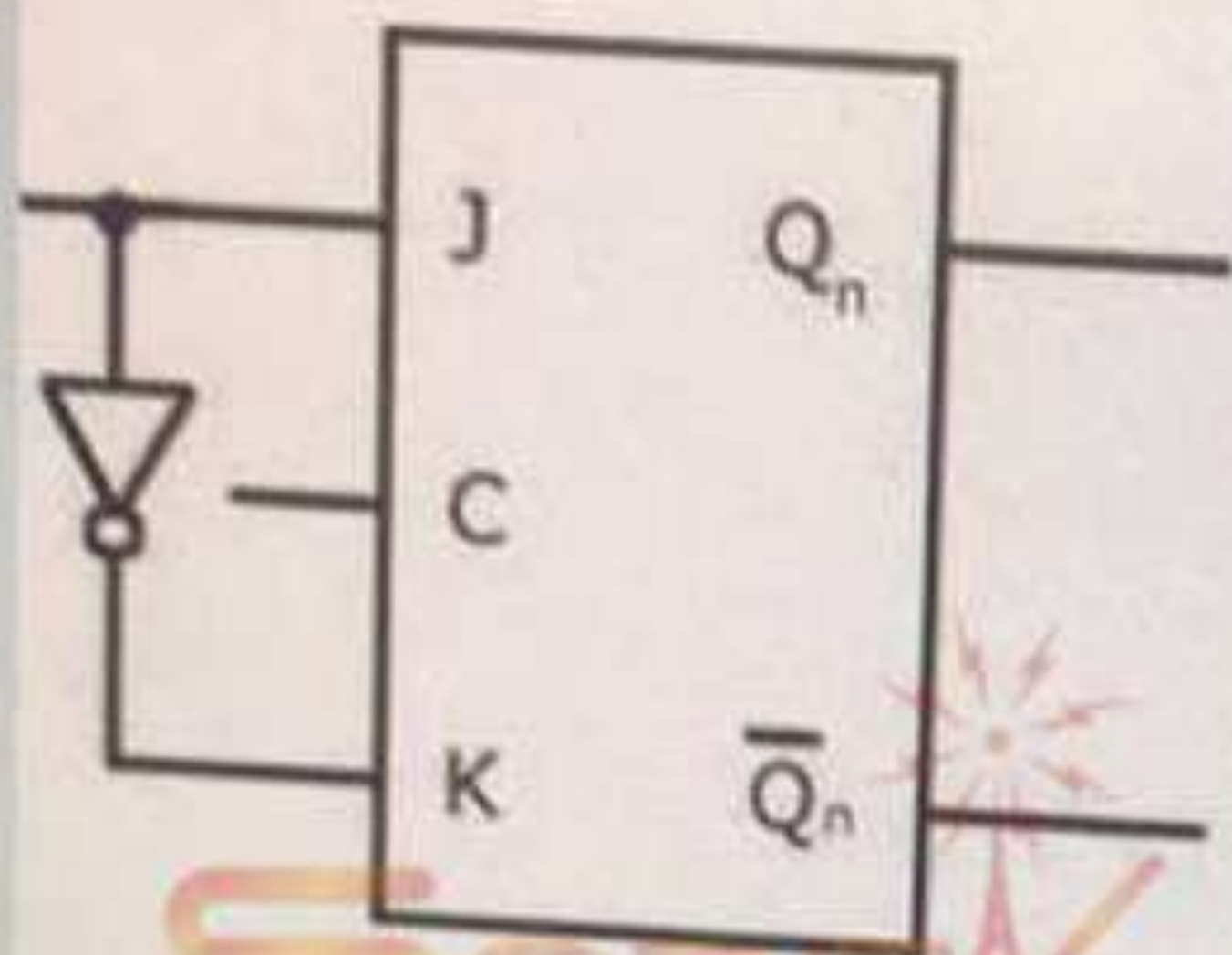
☐ a.
None of these

☐ b.
 $\Sigma(6, 7)$

☒ c.
 $\Sigma(1, 4, 5)$

☐ d.
 $\Sigma(0, 1, 3, 5)$

what kind of flip flop is shown



Select one:

- ☐ a. JK-FF
- ☐ b. SR-FF
- ☐ c. D-FF
- ☐ d. T-FF

A multiplexor with a 4-bit data select input is a

Select one:

- ☐ a.
2:1 multiplexor
- ☐ b.
8:1 multiplexor
- ☐ c.
16:1 multiplexor
- ☒ d.
4:1 multiplexor



In the following truth table, $V = 1$ if and only if the input is valid. What function does the truth table represent?

Inputs				Outputs		
D_0	D_1	D_2	D_3	X_0	X_1	V
0	0	0	0	x	x	0
1	0	0	0	0	0	1
x	1	0	0	0	1	1
x	x	1	0	1	0	1
x	x	x	1	1	1	1

Select one:

- ☐ Multiplexer
- ☐ Decoder
- ☐ Demultiplexer
- ☐ Priority encoder

Which of the following function implements the Karnaugh map shown below?

CD \ AB	00	01	11	10
00	0	0	1	0
01	X	X	1	X
11	0	1	1	0
10	0	1	1	0

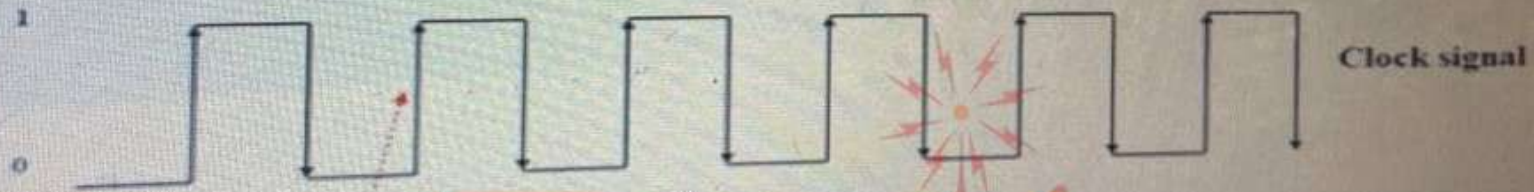
- a. $\bar{A}B + CD$
b. $D(C + A)$
c. $AD + \bar{A}B$
d. $(C + D)(\bar{C} + D)(A + B)$

Select one:

- ☐ a. A
☐ b. B
☐ c. D

☒ d. C Protected with free version of Watermarkly. Full version doesn't put this mark.

The arrow is pointing at the falling edge



Select one:

- ☐ a. true
- ☐ b. false

A multiplexor with a 4-bit data select input is a

Select one:

- ☐ a.
16:1 multiplexor
- ☐ b.
4:1 multiplexor
- ☐ c.
8:1 multiplexor
- ☐ d.
2:1 multiplexor



Which one is a basic comparator?

Select one:

☐ XNOR

☐ AND

☐ NAND

☐ XOR

A flip-flop is edge-sensitive

Select one:

☐ a. false

☐ b. true

The simplified SOP (Sum Of Product) form of the boolean expression $(P + Q' + R') \cdot (P + Q' + R) \cdot (P + Q + R')$ is

Select one:



$(P \cdot Q + R)$



$(P' \cdot Q + R)$



$(P' \cdot Q + R')$



$(P + Q' \cdot R)$



A flip-flop is edge-sensitive

Select one:

☐ a. false

☐ b. true

00

1

x

x

1

01

x

1

11

10

1

x

Select one:



$$b'd' + b'c' + c'd'$$



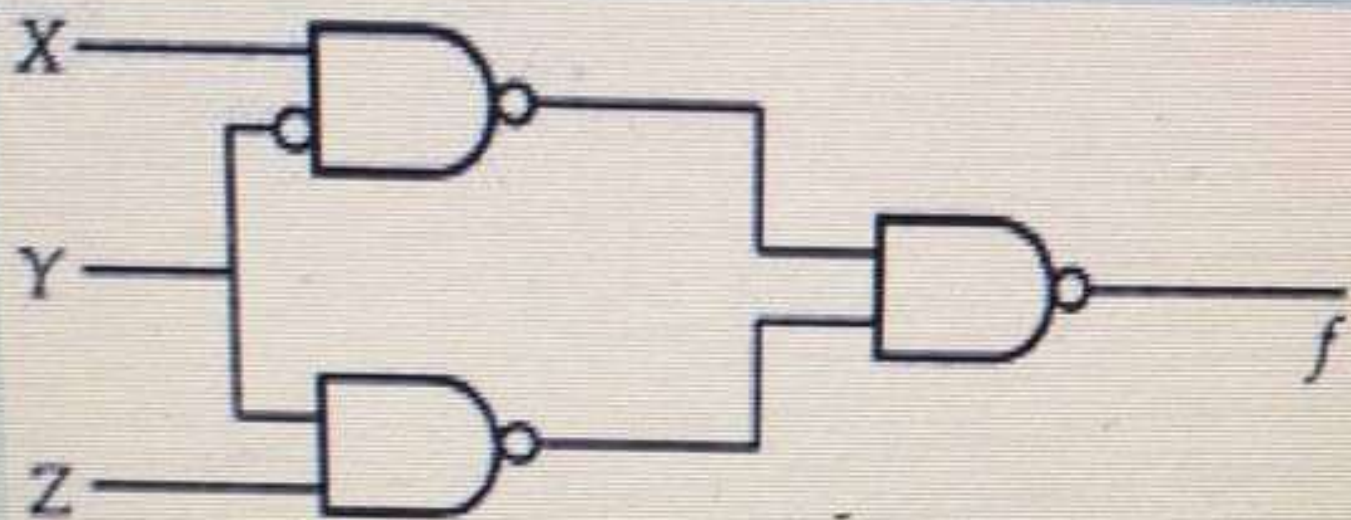
$$b'd$$



$$b'd' + a'b'c'd'$$



$$b'd' + b'c'$$



Which one of the following is TRUE?

Select one:

- ☐ a. f is independent of X
- ☐ b. f is independent of Z
- ☐ c. f is independent of Y
- ☐ d. None of X, Y, Z is redundant

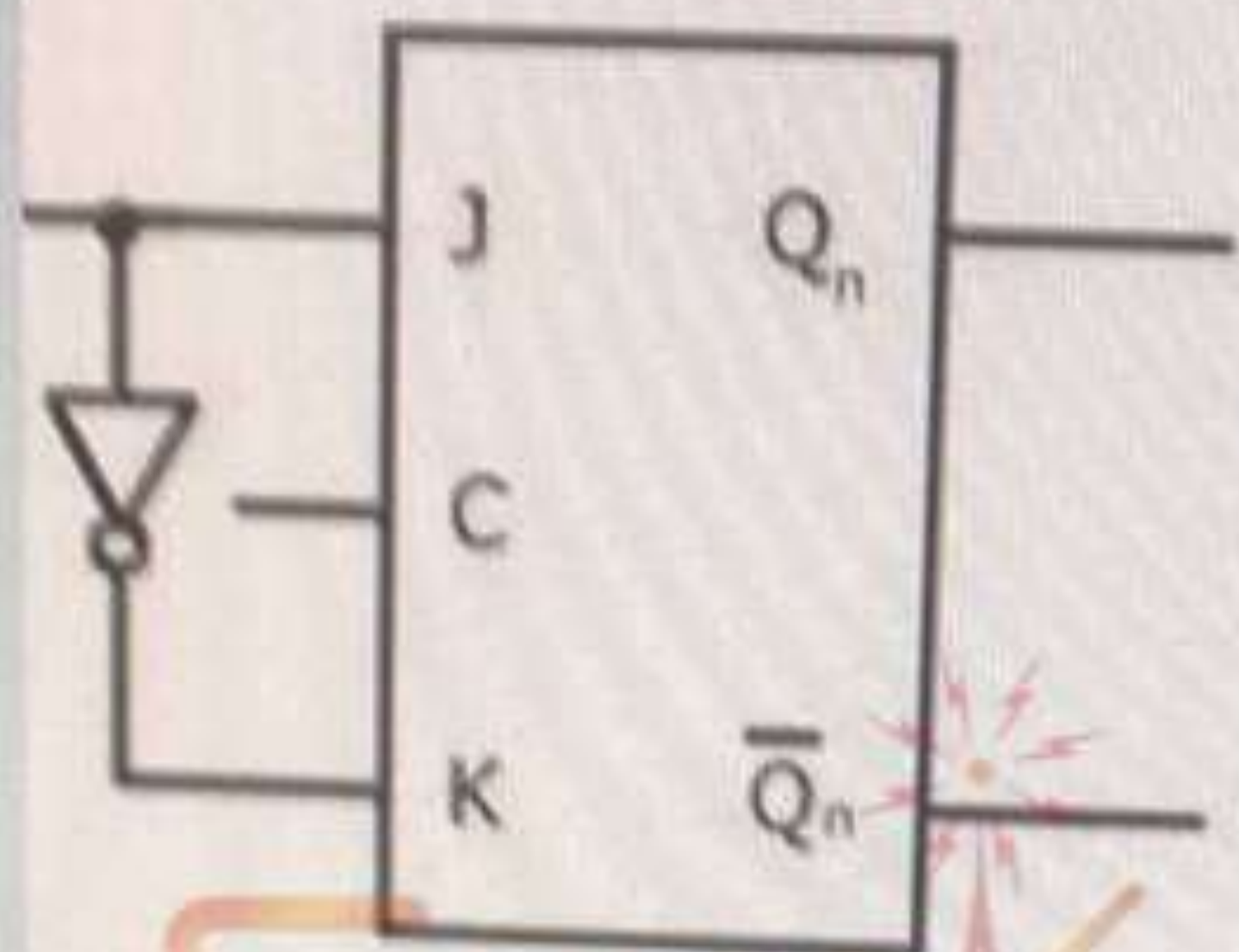
A flip-flop is edge-sensitive

Select one:

☐ a. false

☐ b. true

what kind of flip flop is shown



Select one:

- ☐ a. JK-FF
- ☐ b. SR-FF
- ☐ c. D-FF
- ☐ d. T-FF

Which of the following is Known as universal gates.

Select one:

- ☐ NOR and NAND
- ☐ OR and AND
- ☐ AND and NAND
- ☐ OR and NOR

The minterm expansion of $f(P, Q, R) = PQ + QR' + PR'$ is

Select one:



$m_2 + m_3 + m_4 + m_5$



$m_0 + m_1 + m_6 + m_7$

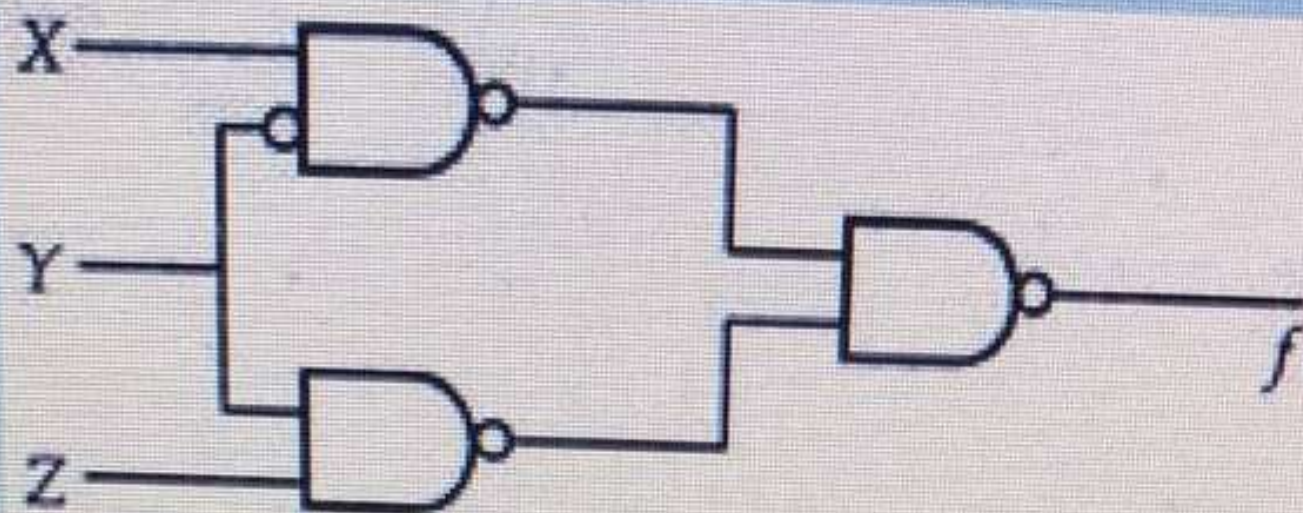


$m_0 + m_1 + m_3 + m_5$



$m_2 + m_4 + m_6 + m_7$





Which one of the following is TRUE?

Select one:

☐ a. f is independent of X

☐ b. f is independent of Z

☐ c. f is independent of Y

☐ d. None of X, Y, Z is redundant

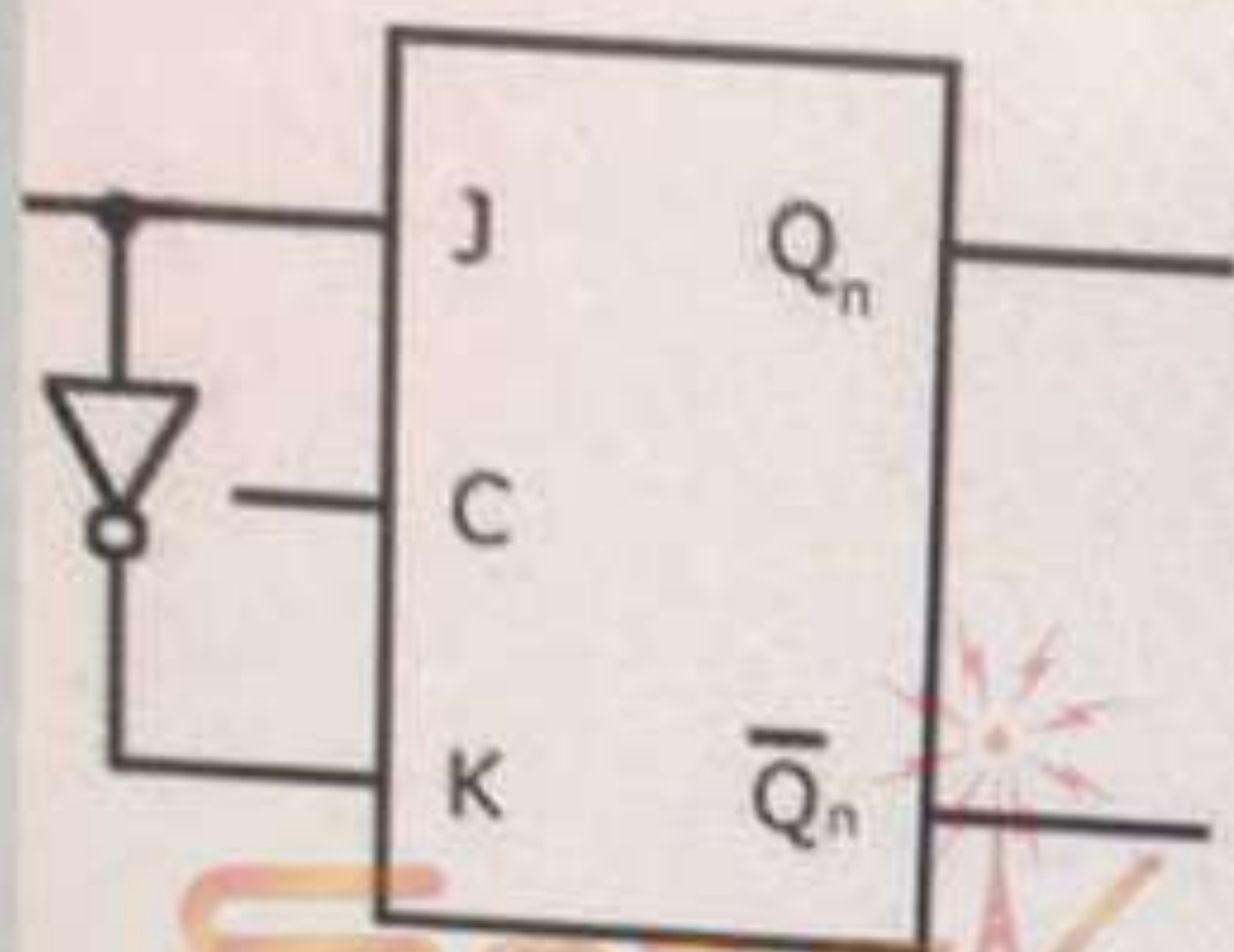
2. There are _____ cells in a 3-variable K-map.

Select one:

- ☐ a. 8
- ☐ b. 12
- ☐ c. 18
- ☒ d. 16



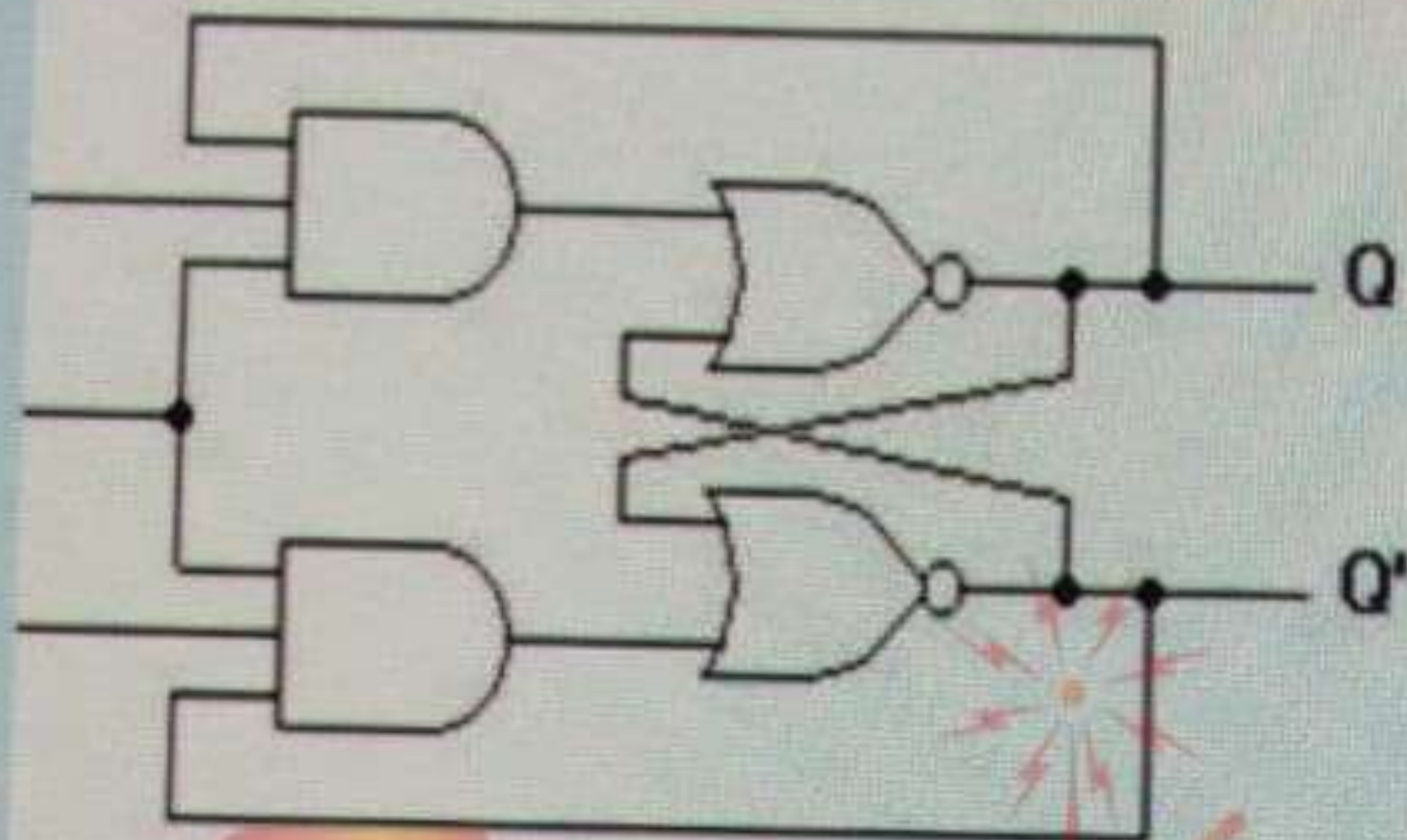
what kind of flip flop is shown



Select one:

- ☐ a. JK-FF
- ☐ b. SR-FF
- ☐ c. D-FF
- ☐ d. T-FF

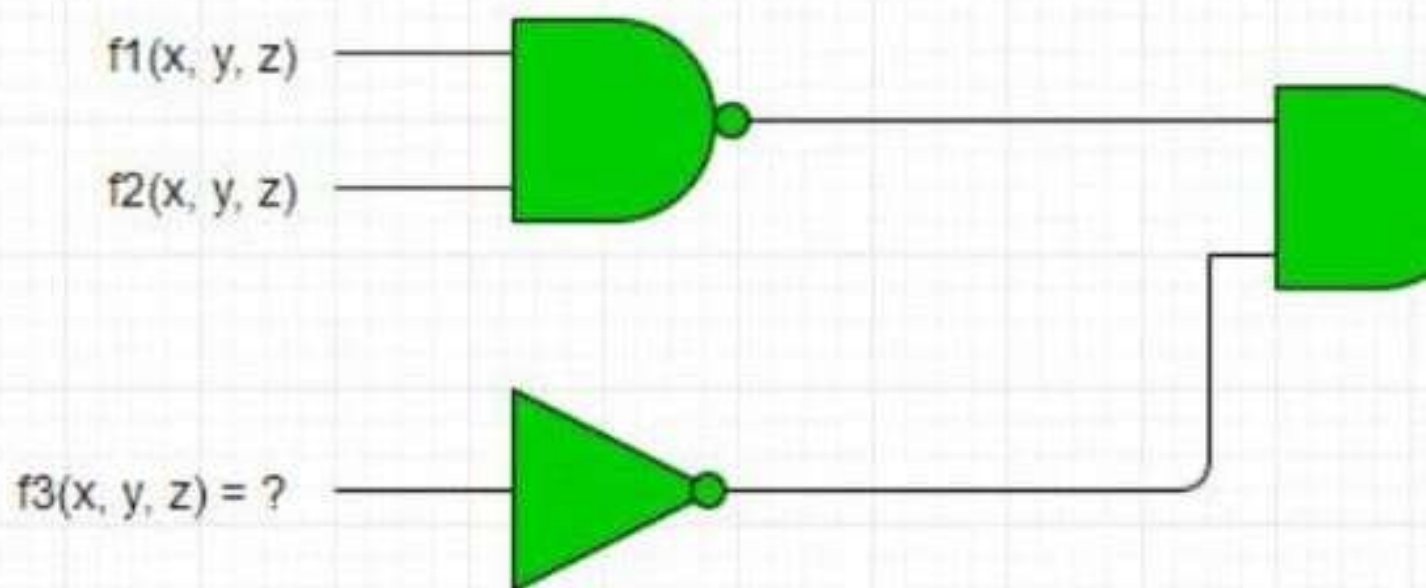
what kind of flip flop is shown



Select one: YTECHNIC

- ☐ a. T-FF
- ☐ b. SR-FF
- ☐ c. D-FF
- ☐ d. JK-FF

C



Given that

$$f1(x, y, z) = \sum(0, 1, 3, 5),$$

$$f2(x, y, z) = \sum(6, 7) \text{ and}$$

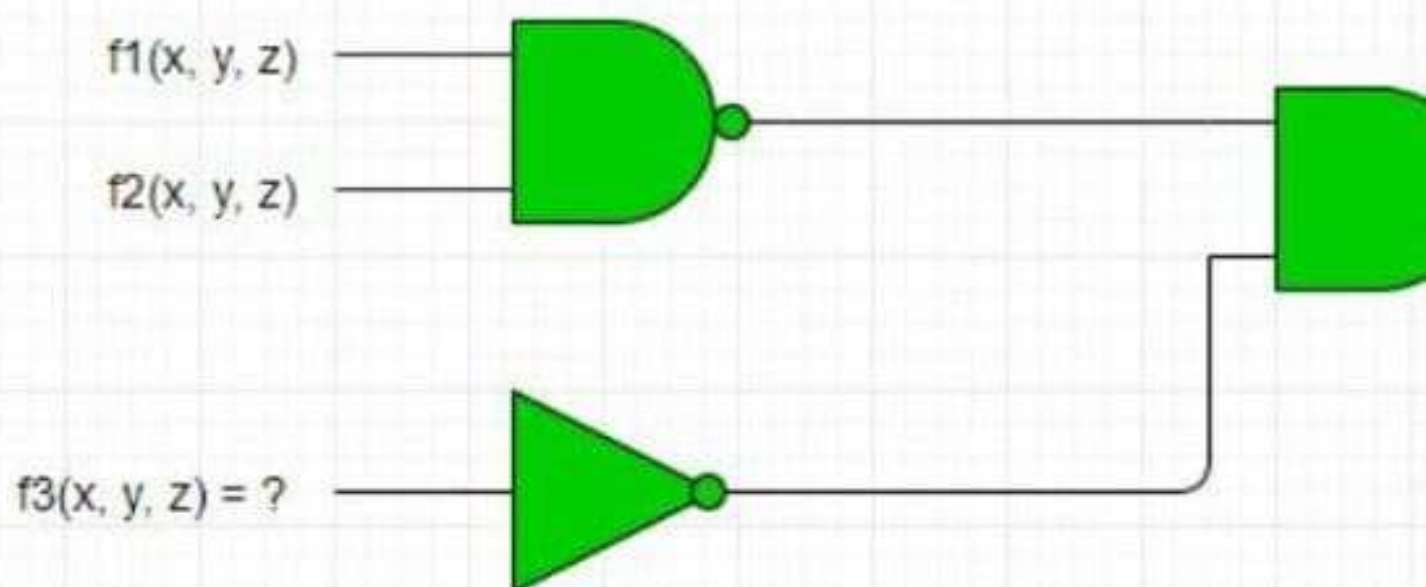
$$f3(x, y, z) = \sum(1, 4, 5),$$

$f3$ is:

Select one:

- ☐ a.
 $\sum(1, 4, 5)$
- ☐ b.
 $\sum(6, 7)$
- ☐ c.
None of these
- ☐ d.
 $\sum(0, 1, 3, 5)$

C



Given that

$$f1(x, y, z) = \sum(0, 1, 3, 5),$$

$$f2(x, y, z) = \sum(6, 7) \text{ and}$$

$$f(x, y, z) = \sum(1, 4, 5),$$

$f3$ is:

Select one:

- ☐ a. $\sum(1, 4, 5)$
- ☐ b. $\sum(6, 7)$
- ☐ c. None of these
- ☐ d. $\sum(0, 1, 3, 5)$

cd \ ab	00	01	11	10
00	1	x	x	1
01	x			1
11				
10	1			x

Select one:



$$b'd' + b'c' + c'd'$$



$$b'd$$



$$b'd' + a'b'c'd'$$



$$b'd' + b'c$$

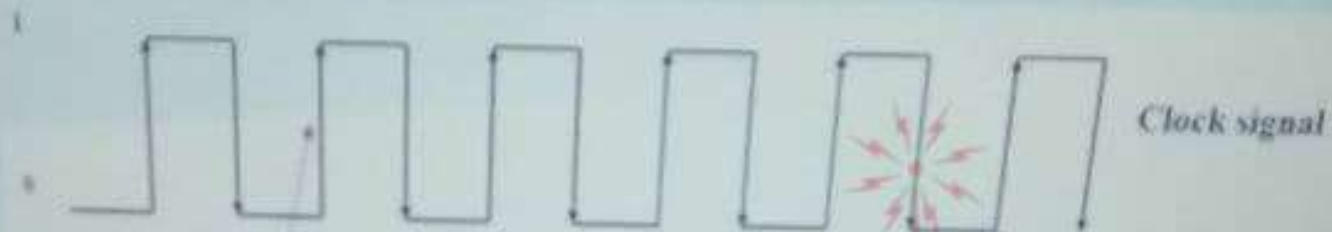
In a 7-segment display decoder driver using positive logic (common cathode display), a low output level represents a segment in the ON state.

Select one:

- ☐ a. TRUE
- ☐ b. FALSE



The arrow is pointing at the falling edge



Spark
POLYTECHNIC

المزاج الفطري

a true ☐

is false ☐

fill the state table

JKQ Q+

000 ?????



Select one:

☒ a. Q

☐ b. Q'

☐ c. 1

☐ d. 0

fill the state table

JKQ Q+

1 1 0 ? ? ? ?



Select one:

- ☐ a. Q
- ☐ b. Q'
- ☐ c. 1
- ☐ d. 0

fill the state table

JKQQ+

1 1 1 ? ? ? ?



Select one:

☐ a. Q'

☐ b. Q

☒ c. 0

☐ d. 1

fill the state table

JK Q Q+

1 1 1 ? ? ? ?



Select one:

☐ a. Q'

☐ b. Q

☒ c. 0

☐ d. 1

fill the state table

JKQQ+

110????



Select one:

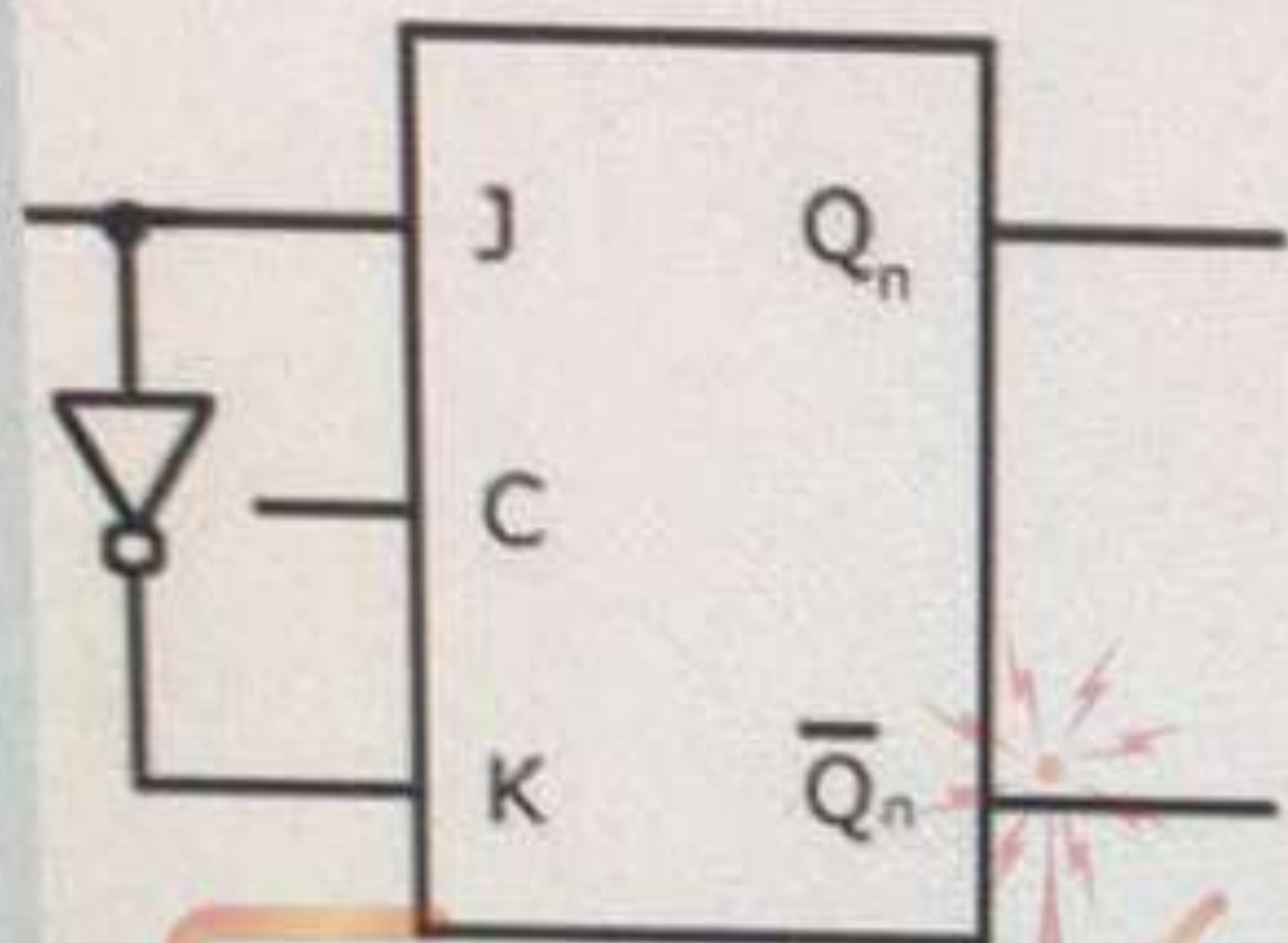
☐ a. 1

☐ b. 0

☐ c. Q'

☐ d. Q

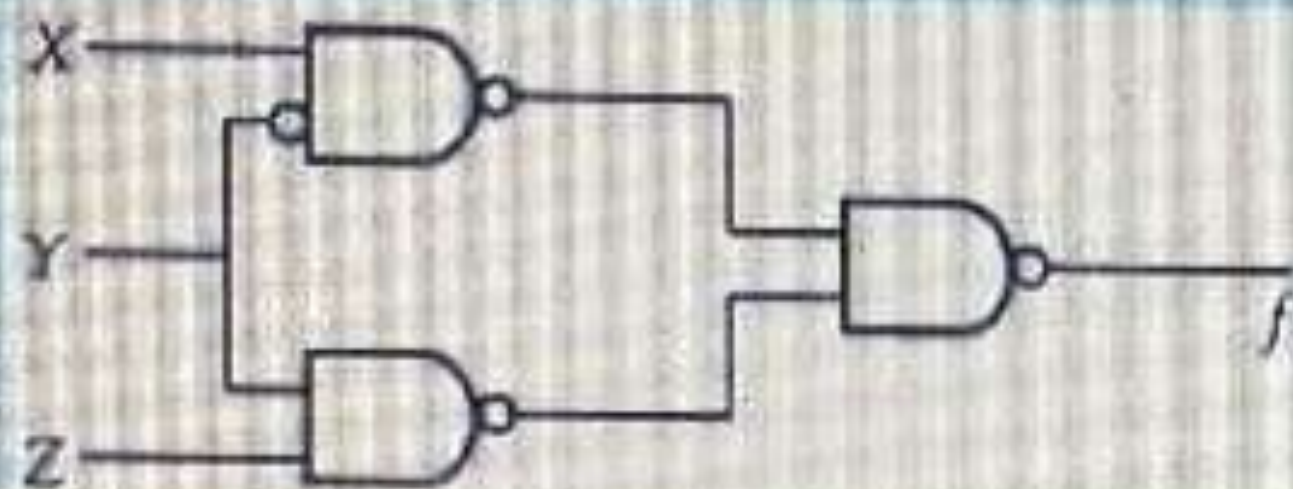
what kind of flip flop is shown



Sparx
POLYTECHNIC

Select one:

- ☐ a. JK-FF
- ☐ b. SR-FF
- ☐ c. D-FF
- ☐ d. T-FF



Which one of the following is TRUE?

Select one:



a.

None of X, Y, Z is redundant



b.

f is independent of Z



c.

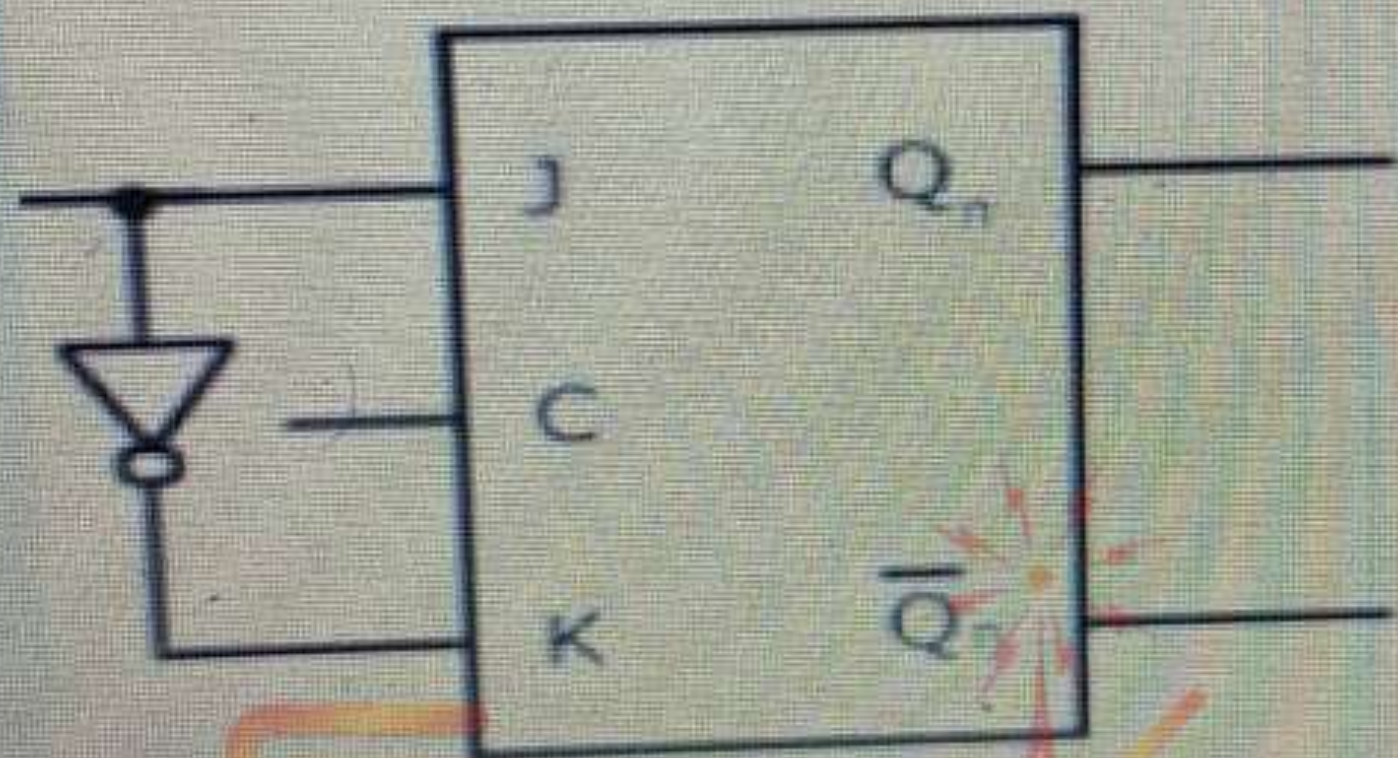
f is independent of Y



d.

f is independent of X

what kind of flip flop is shown



Select one:

- ☐ a. D-FF
- ☐ b. T-FF
- ☐ c. JK-FF
- ☐ d. SR-FF

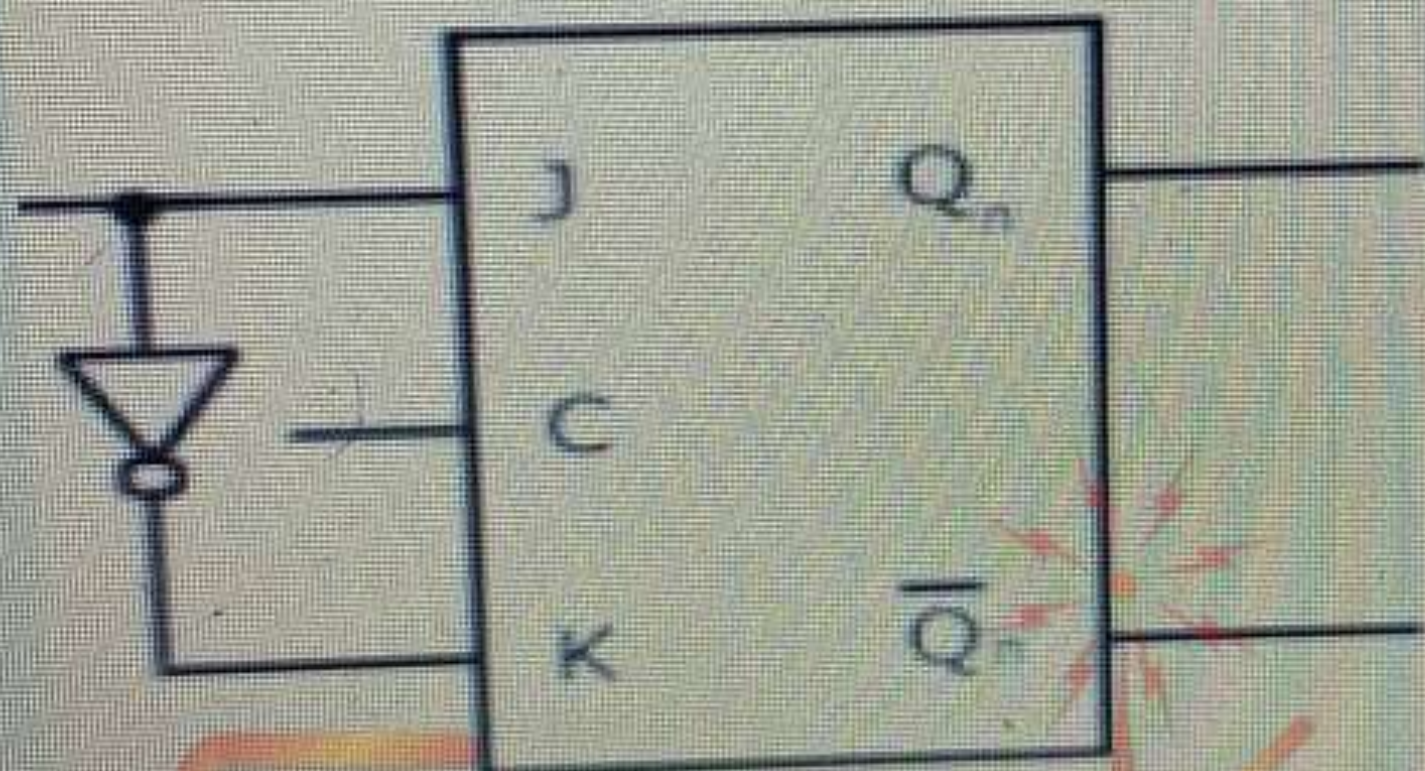
A latch is edge sensitive

Select one:

- ☐ a. true
- ☐ b. false



what kind of flip flop is shown



Select one:

- ☐ a. D-FF
- ☐ b. T-FF
- ☐ c. JK-FF
- ☐ d. SR-FF

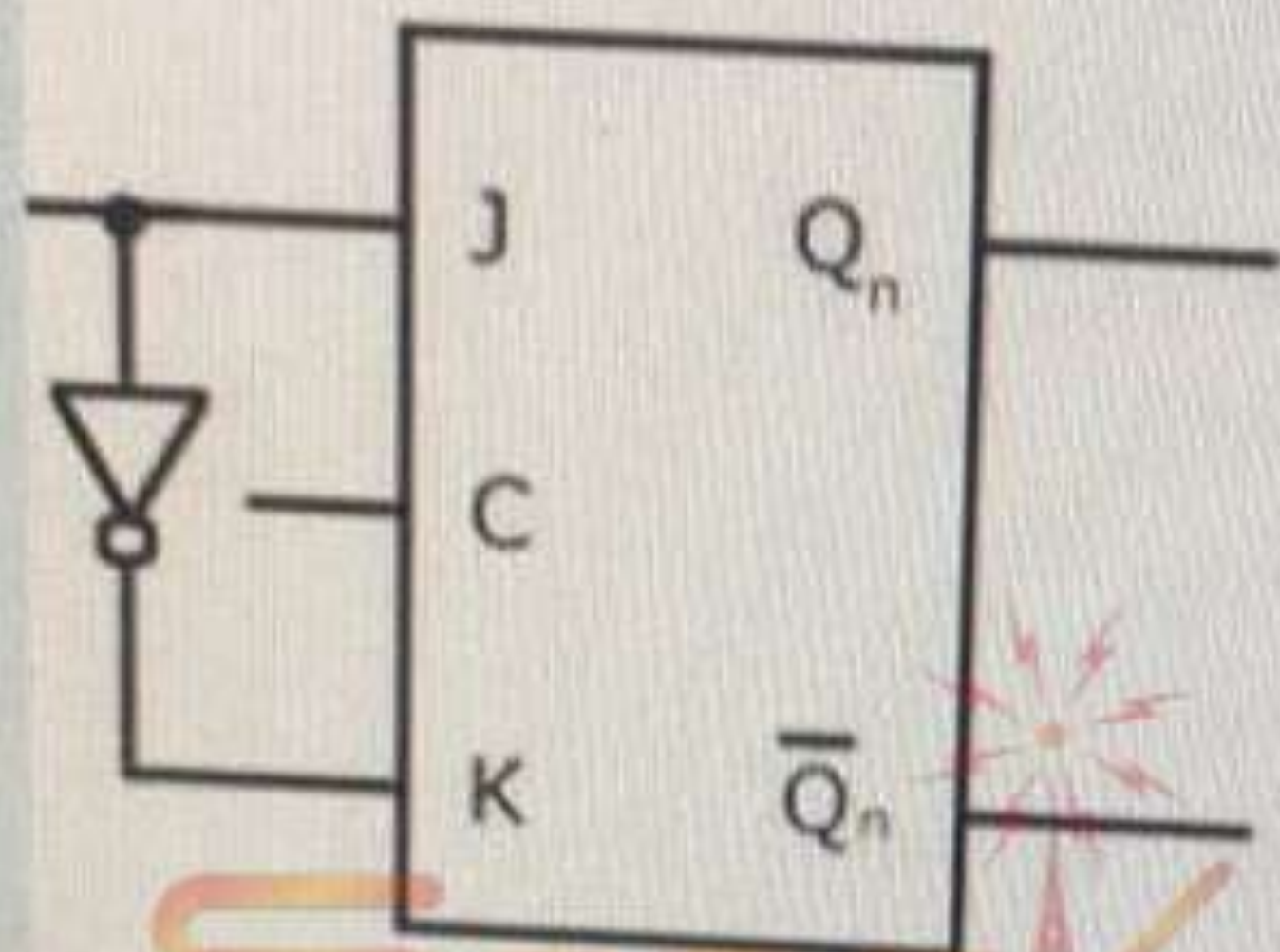
If A and B are the inputs of a half adder, the carry is given by _____

Select one:

- ☐ a. $A \text{ XOR } B$
- ☐ b. $A \text{ EX-NOR } B$
- ☐ c. $A \text{ AND } B$
- ☐ d. $A \text{ OR } B$



what kind of flip flop is shown



Select one:

- ☐ a. JK-FF
- ☐ b. SR-FF
- ☐ c. D-FF
- ☐ d. T-FF

DIGITAL

A latch is edge sensitive

Select one:

- ☐ a. true
- ☐ b. false





Select one:

☐ a.

$$w'x + y'z + xy$$

☐ b.



☐ c.

$$xy + y'z$$

☐ d.

$$wx'y' + xy + xz$$

C

Given that

$$f_1(x, y, z) = \Sigma(0, 1, 3, 5),$$
$$f_2(x, y, z) = \Sigma(6, 7) \text{ and}$$
$$f_3(x, y, z) = \Sigma(1, 4, 5),$$

f_3 is:

Select one:

- ☐ a.
 $\Sigma(1, 4, 5)$
- ☐ b.
 $\Sigma(6, 7)$
- ☐ c.
None of these
- ☐ d.
 $\Sigma(0, 1, 3, 5)$



Select one:

☐ a.

$$w'x + y'z + xy$$

☐ b.

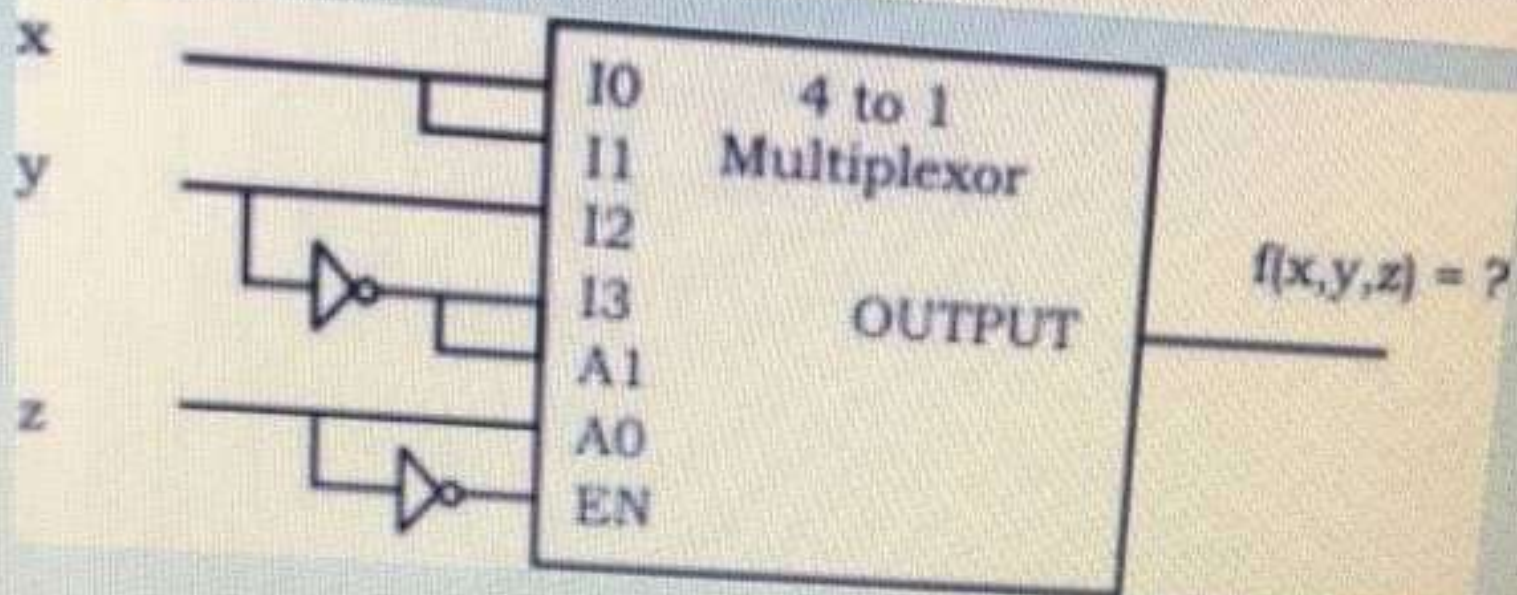
$$xz + y$$

☐ c.

$$xy + y'z$$

☐ d.

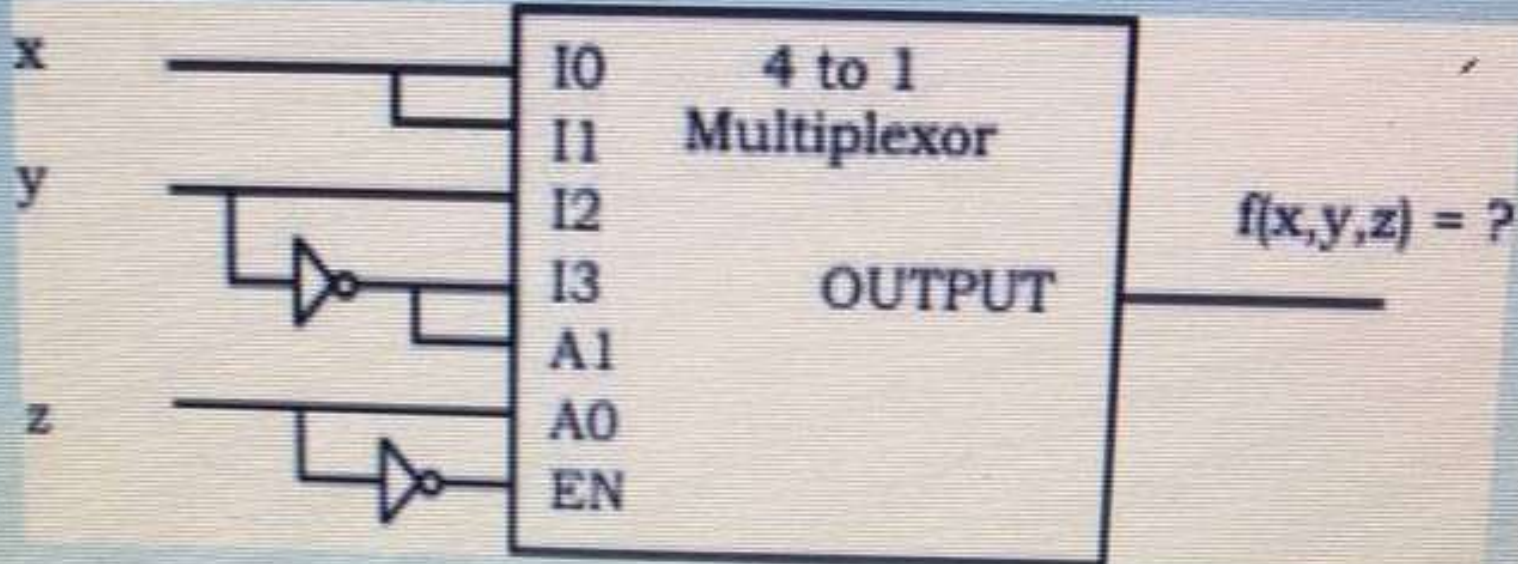
$$wx'y' + xy + xz$$



F=

Select one:

- ☐ a. $xy + z$
- ☐ b. $x + z$
- ☐ c. None of these
- ☒ d. xyz'



F=

Select one:

☐ a. $x + z$

☐ b. $xy + z$

☐ c. None of these

☐ d. xyz'

A multiplexor with a 4-bit data select input is a

Select one:

- ☐ a.
2:1 multiplexor
- ☐ b.
8:1 multiplexor
- ☐ c.
16:1 multiplexor
- ☒ d.
4:1 multiplexor

POLYTECHNIC



A multiplexor with a 4-bit data select input is a

Select one:

- ☐ a.
2:1 multiplexor
- ☐ b.
8:1 multiplexor
- ☐ c.
16:1 multiplexor
- ☒ d.
4:1 multiplexor

POLYTECHNIC



Which of following are known as universal gates?

Select one:

- ☐ a. NAND & NOR
- ☐ b. AND & OR
- ☐ c. XOR & OR
- ☐ d. EX-NOR & XOR



If you want to build 6 bit parallel adder subtractor , you need :

Select one:

- ☐ a. 6 half adders
- ☒ b. 12 half adders
- ☐ c. 7 half adders
- ☐ d. 18 half adders



The function Y for the following block diagram represents:



Select one:

- ☐ a. AND
- ☒ b. INVERTER
- ☐ c. XNOR
- ☐ d. BUFFER

what kind of FF does the table represent?

INPUT

Q+

0	0	Q
0	1	0
1	0	1
1	1	?

Select one:

☒ SR-FF

☐ T-FF

☐ D-FF

☐ NONE

☐ JK-FF

If Figure (1) is equivalent to Figure (2), and B in Figure (1) is used as selector. What is the value of (?)

FIGURE (1)

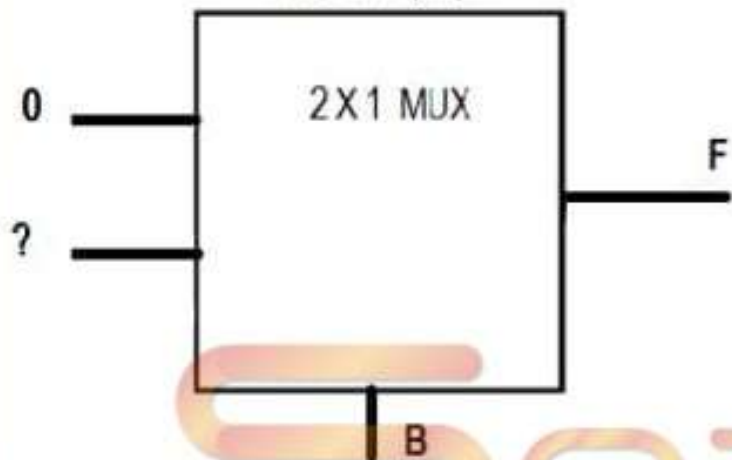
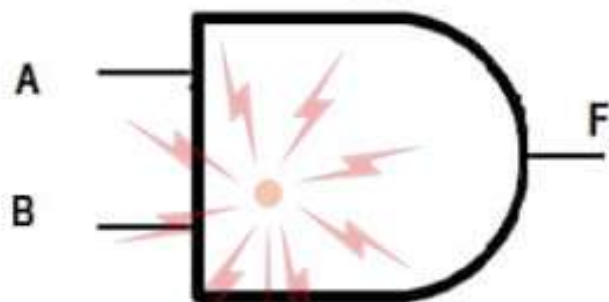


FIGURE (2)



Select one:

- ☐ a. 1
- ☒ b. A
- ☐ c. B'
- ☐ d. A'

For the given state table below using row match principle. Which one is reduced state table?

Present State	Next State		Output	
	X=0	X=1	X=0	X=1
H	H	J	0	0
J	K	M	0	1
K	K	L	0	0
L	H	J	0	0
M	K	F	0	1
N	H	J	0	0

Select one:

Present State	Next State		Output	
	X=0	X=1	X=0	X=1
H	M	J	0	0
J	K	M	0	1
K	M	H	0	0
M	K	M	0	1

☒ a.

Present State	Next State		Output	
	X=0	X=1	X=0	X=1
H	H	J	0	0
J	K	M	0	1
K	K	L	0	0
L	H	J	0	0
M	K	F	0	1
N	H	J	0	0

Select one:

☐ a.

Present State	Next State		Output	
	X=0	X=1	X=0	X=1
H	M	J	0	0
J	K	M	0	1
K	M	H	0	0
M	K	M	0	1

☒ b.

Present State	Next State		Output	
	X=0	X=1	X=0	X=1
H	H	J	0	0
J	K	M	0	1
K	K	H	0	0
M	K	H	0	1

☐ c.

Present State	Next State		Output	
	X=0	X=1	X=0	X=1
H	H	J	0	0
J	K	M	0	1
K	K	H	0	0
M	K	N	0	1
N	H	J	0	0

what kind of FF does the table represent?

INPUT

Q+

0	0	Q
0	1	0
1	0	1
1	1	Q'

Select one:

- ☐ SR-FF
- ☐ NONE
- ☐ D-FF
- ☒ JK-FF
- ☐ T-FF

Design a sequence detector to detect the sequence pattern 1110

[x is the input] , [A,B is present state] , [A+,B+ is next state]

ABX	A+	B+
000	0 ▾	0 ▾
001	0 ▾	1 ▾
010	0 ▾	0 ▾
011	1 ▾	1 ▾
100	0 ▾	0 ▾
101	0 ▾	1 ▾
110	1 ▾	0 ▾
111	1 ▾	1 ▾



What are the outputs of a four-bit magnitude comparator when the inputs are $A = 0110$ and $B = 1010$?

Select one:

- ☐ a. $A < B$ is 1 , $A = B$ is 1 , $A > B$ is 0
- ☐ b. $A < B$ is 0 , $A = B$ is 0 , $A > B$ is 1
- ☐ c. $A < B$ is 1 , $A = B$ is 0 , $A > B$ is 0
- ☒ d. $A < B$ is 0 , $A = B$ is 1 , $A > B$ is 0



How many 3-to-8 line decoders with an enable input are needed to construct a 6-to-64 line decoder without using any other logic gates?

Select one:

- ☒ a. 10
- ☐ b. 7
- ☐ c. 8
- ☐ d. 9



The group of bits 10110111 is serially shifted (right-most bit first) into an 8-bit parallel output shift register with an initial state 11110000. After two clock pulses, the register contains

Select one:

- ☐ 10111000
- ☒ 11111100
- ☐ 11110000
- ☐ 10110111



When Design a counter that counts in the sequence XY : 11 $\rightarrow$ 01 $\rightarrow$ 10 $\rightarrow$ 00 and repeat , using T flip flops.(X is the MSB)

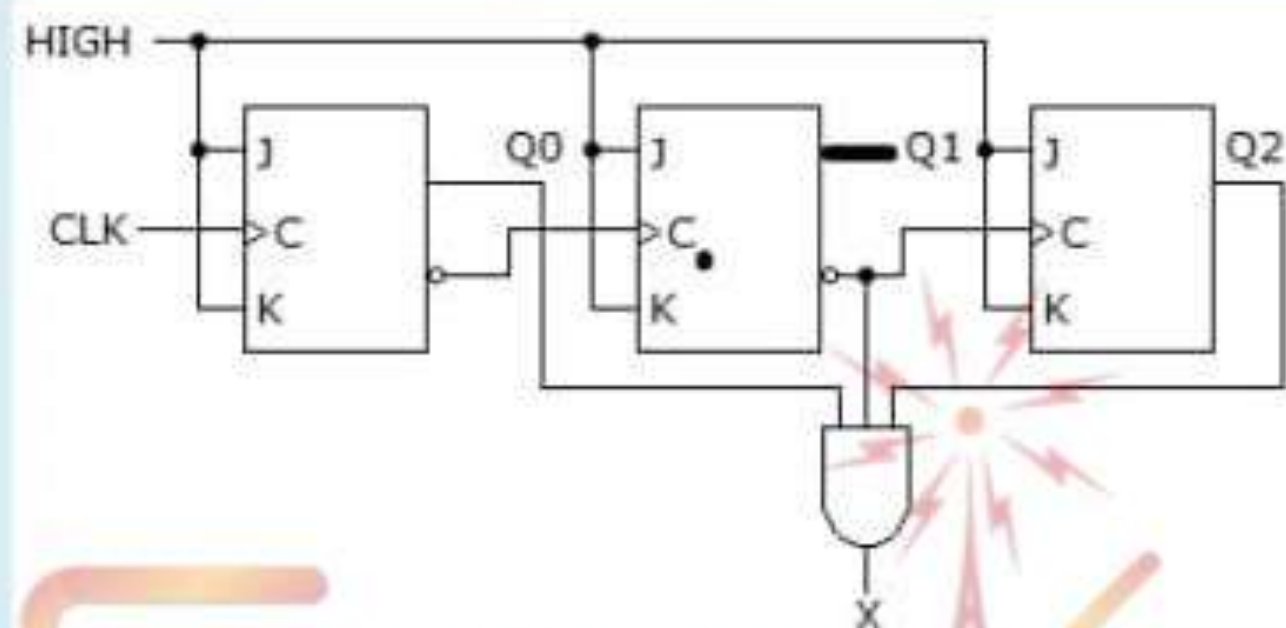
Answer the following questions

1. The boolean function for T_X is

2. The boolean function for T_Y is

3. How many T Flip Flop you must use to design the circuit?

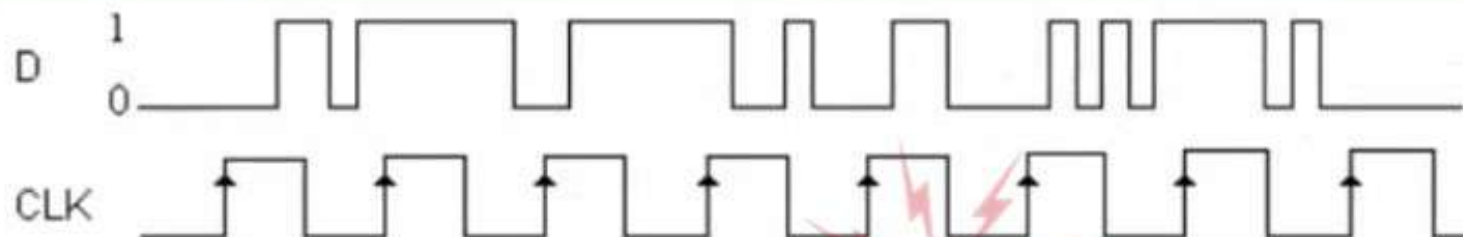
What decimal value is required to make $X=1$?



Select one:

- ☐ a. 5
- ☒ b. 1 or 4
- ☐ c. 7
- ☐ d. 1

For the following D flip-flop Which of the diagrams is representing Q:



Select one:

☐ a.



☒ b.



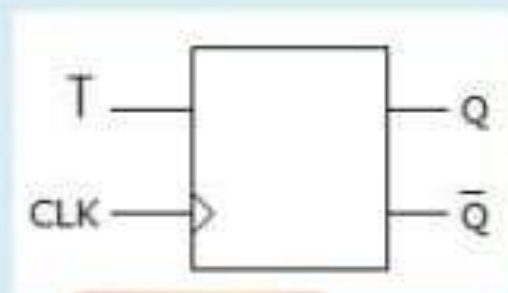
☐ c.



☐ d.



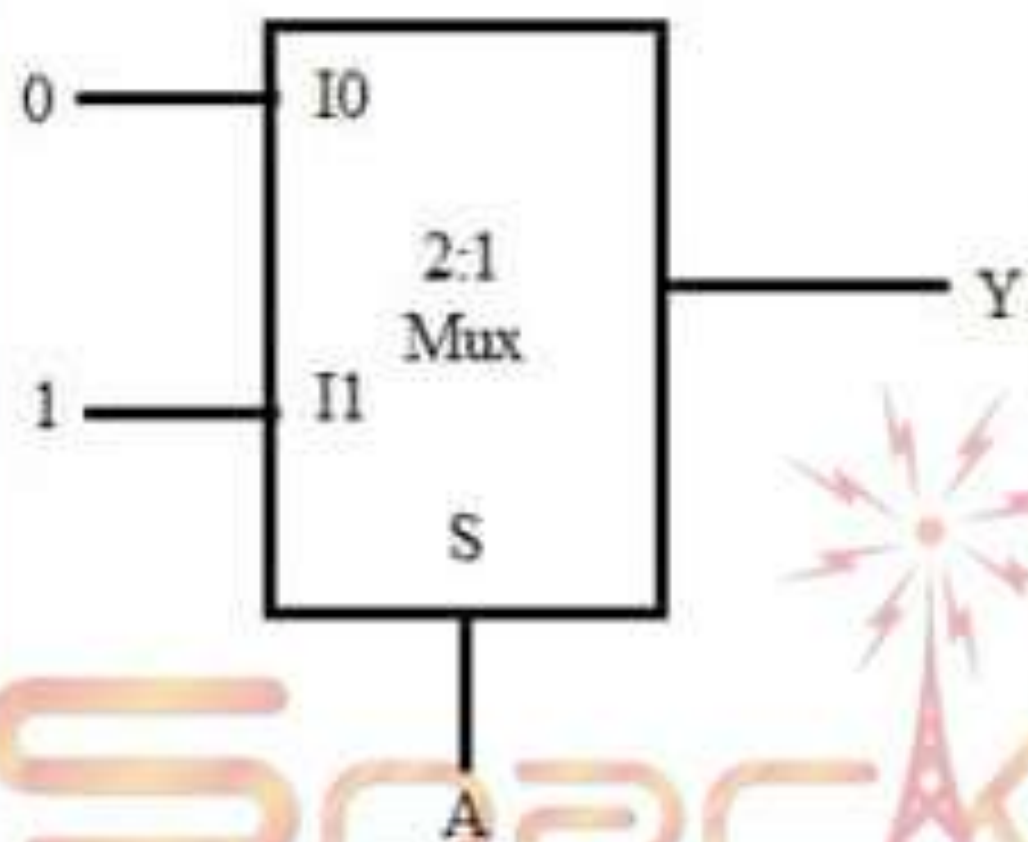
The symbols on this flip-flop device indicate _____.



Select one:

- ☐ a. triggering takes place on the negative-going edge of the CLK pulse
- ☐ b. triggering can take place anytime during the LOW level of the CLK waveform
- ☐ c. triggering can take place anytime during the HIGH level of the CLK waveform
- ☒ d. triggering takes place on the positive-going edge of the CLK pulse

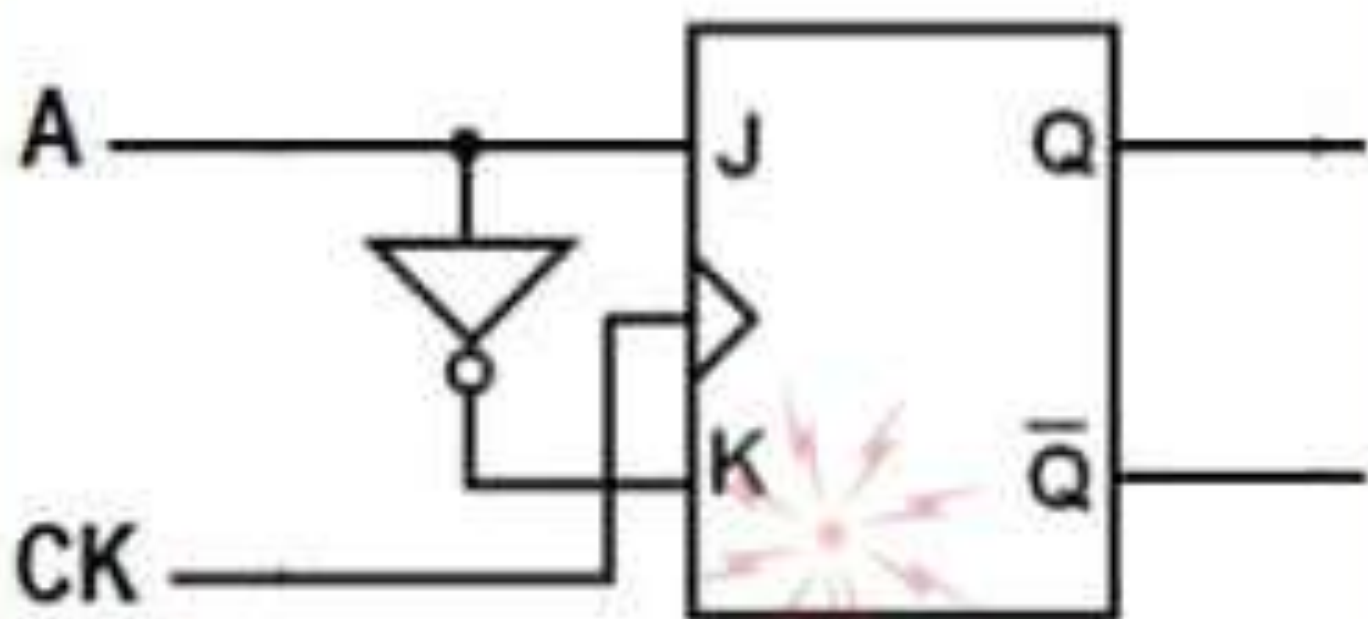
The function Y for the following block diagram represents



Select one:

- ☒ a. XOR
- ☐ b. AND
- ☐ c. BUFFER
- ☐ d. INVERTER

The Figure shown below works as :



Select one:

- ☒ a. D flip flop
- ☐ b. 3x8 Decoder
- ☐ c. S R flip flop
- ☐ d. T flip flop

Which of the following function implements the Karnaugh map shown below?

CD \ AB	00	01	11	10
00	0	0	1	0
01	X	X	1	X
11	0	1	1	0
10	0	1	1	0

a. $\bar{A}B + CD$

b. $D(C + A)$

c. $AD + \bar{A}B$

d. $(C + D)(\bar{C} + D)(A + B)$

Select one:

☐ a. C

☒ b. B

☐ c. D

☐ d. A

fill the state table

JK Q Q+

1 1 0 ? ? ? ?



Select one:

- ☐ a. 1
- ☐ b. Q'
- ☐ c. Q
- ☐ d. 0

In the following truth table, $V = 1$ if and only if the input is valid. What function does the truth table represent?

Inputs				Outputs		
D_0	D_1	D_2	D_3	X_0	X_1	V
0	0	0	0	x	x	0
1	0	0	0	0	0	1
x	1	0	0	0	1	1
x	x	1	0	1	0	1
x	x	x	1	1	1	1

Select one:

- ☐ Priority encoder
- ☐ Demultiplexer
- ☐ Multiplexer
- ☐ Decoder

A demultiplexer is used to

Select one:

- ☐ a. Perform serial to parallel conversion
- ☐ b. Select data from several inputs and route it to single output
- ☒ c. Route the data from single input to one of many outputs
- ☐ d. Both route the data and perform serial to parallel

The correct answer is: Both route the data and perform serial to parallel

- ☐ b. to convert a binary number into decimal one
- ☐ c. to input data in binary way into a digital system
- ☐ d. to input data in binary way into a Analog system

Question 14

Not yet answered

Marked out of 1.00

Flag question

A binary decimal coder is composed of:

Select one:

- ☐ a. logic gates only
- ☐ b. none of them
- ☐ c. logic gates and flip-flops
- ☐ d. flip-flops only

Question 15

Not yet answered

Marked out of 1.00

Flag question

How many inputs and outputs has a decimal to BCD encoder without priority?

Select one:

- ☐ a. 10 inputs and 3 outputs
- ☐ b. 9 inputs and 3 outputs
- ☐ c. 9 or 10 inputs and 4 outputs

If the input J is connected through K input of J-K, then flip-flop will behave as a

Select one:

- ☐ a. S-R flip-flop
- ☐ b. T type flip-flop
- ☐ c. Master slave JK flip-flop
- ☒ d. D type flip-flop ✖

The correct answer is: T type flip-flop

- ☐ a. a power transistor
- ☐ c. AND or OR gates
- ☐ d. clock pulse generators

Question 7

Not yet answered

Marked out of 1.00

Flag question

What is the function of a decimal-binary encoder?

Select one:

- ☐ a. to convert from binary to decimal code
- ☒ b. to convert from decimal to binary code
- ☐ c. to convert a BCD number into an hexadecimal one
- ☐ d. to convert an analog signal into a digital one

Question 8

Not yet answered

Marked out of 1.00

Flag question

In a 7-segment display decoder driver using negative logic (common anode display), a low output level

Select one:

- ☐ a. the question wrong

- ☐ c. to convert a BCD number into an hexadecimal one
- ☐ d. to convert an analog signal into a digital one

Question 8

Not yet answered

Marked out of 1.00

Flag question

In a 7-segment display decoder driver using negative logic (common anode display), a low output level represents a segment in the ON state.

Select one:

- ☐ a. the question wrong
- ☐ b. True
- ☒ c. False
- ☐ d. No answer

Question 9

Not yet answered

Marked out of 1.00

Flag question

When happens a state change in a clocked sequential circuit?

Select one:

- ☐ a. on the rising or falling edges of the clock
- ☐ b. at any moment, it depends on the inputs
- ☐ c. during the period "T" of the clock
- ☐ d. when the clock signal is high

Question 10

Not yet answered

Marked out of 1.00

Which logic gates can be used to make a flip-flop?

Select one:

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An RS flip-flop is the basic type of sequential logic circuits

How many flip flops we have?

Answer: 4 flip flops



What is the function of a binary-decimal coder?

Select one:

- ☐ a. to input data in binary way into a Analog system
- ☒ b. to convert a binary number into decimal one
- ☐ c. to count
- ☐ d. to input data in binary way into a digital system

☐ d. 1 NAND

Question 11

Not yet answered

Marked out of 1.00

Flag question

Which is the better type of drive for an led display?

Select one:

- ☐ a. Micro
- ☐ b. Current
- ☐ c. R
- ☐ d. Voltage



Question 12

Not yet answered

Marked out of 2.00

Flag question

How many flip flops we have?

Answer: 4

- ☐ c. 7-segment display
- ☐ d. dot-matrix display (e.g. 6 x 8)

What do you mean, when you say that a flip-flop is an indeterminate state?

Select one:

- ☐ a. that the outputs are not at level 0, nor 1
- ☐ b. that the outputs are not fired
- ☐ c. that after switching flip-flop, the memory state is undefined, and can be $Q=0$ and $??=1$ or $Q=1$ and $??=0$
- ☐ d. that the flip-flop tends to oscillate indefinitely

How is numerical data displayed normally, in digital systems?

Select one:

- ☒ a. with a 7-segment display
- ☐ b. with a counter
- ☐ c. with a multimeter

The number of full and half-adders required to add 16-bit numbers is

Select one:

- ☐ a. 16 half-adders, 0 full-adders
- ☐ b. 1 half-adder, 15 full-adders
- ☒ c. 4 half-adders, 12 full-adders 
- ☐ d. half-adders, 8 full-adders

The correct answer is: 1 half-adder, 15 full-adders

- ☐ b. 4 inputs and 9 outputs
- ☒ c. 4 inputs and 10 outputs
- ☐ d. 9 inputs and 4 outputs

Question 3

Not yet answered

Marked out of 1.00

Flag question

Which kind of display is it better to use for the display of the alphanumeric characters?

Select one:

- ☐ a. 14-16-18-segment display
- ☐ b. nixie display
- ☐ c. 7-segment display
- ☐ d. dot-matrix display (e.g. 6 x 8)

Question 4

Not yet answered

Marked out of 1.00

Flag question

What do you mean, when you say that a flip-flop is an indeterminate state?

Select one:

- ☐ a. that the outputs are not at level 0, nor 1
- ☐ b. that the outputs are not fired

Question 7

Not yet answered

Marked out of 1.00

Flag question

Which is the better type of drive for an led display?

Select one

- ☐ a. P
- ☒ b. Current
- ☐ c. Micro
- ☐ d. Voltage

Question 8

Not yet answered

Marked out of 1.00

Flag question

How is numerical data displayed normally, in digital systems?

Select one

- ☒ a. with a counter
- ☐ b. with a multimeter
- ☐ c. with a speaker
- ☐ d. with a 7-segment display

Question 9

Not yet answered

Marked out of 1.00

Flag question

When happens a state changement in a clocked sequential circuit?

Select one

- ☐ a. during the period "T" of the clock
- ☐ b. at any moment it depends on the inputs

- ☐ a. the question wrong
- ☐ b. True
- ☐ c. False
- ☐ d. No answer

Question 9

Not yet answered

Marked out of 1.00

Flag question

When happens a state change in a clocked sequential circuit?

Select one:

- ☐ a. on the rising or falling edges of the clock
- ☐ b. at any moment, it depends on the inputs
- ☐ c. during the period "T" of the clock
- ☐ d. when the clock signal is high

Question 10

Not yet answered

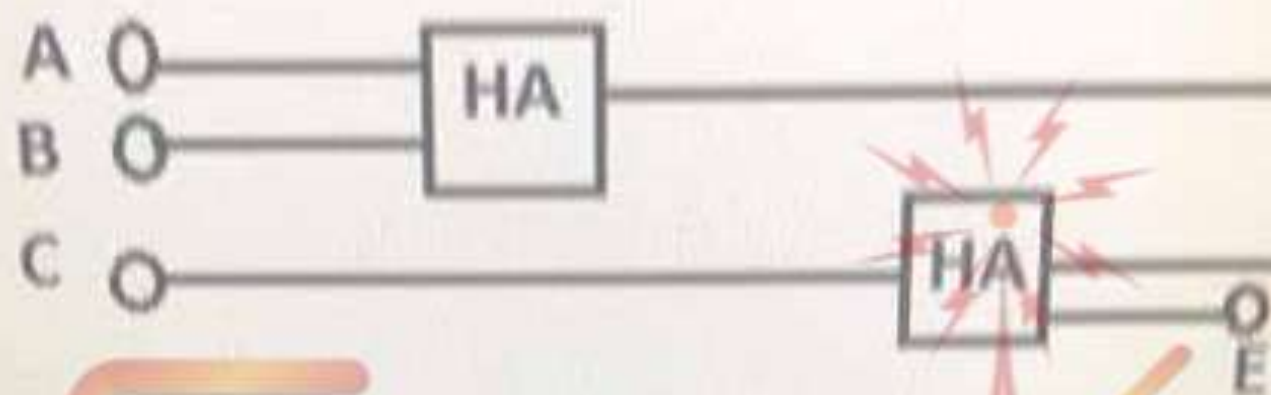
Marked out of 1.00

Which logic gates can be used to make a flip-flop?

Select one:

- ☐ a. EXCLUSIVE OR

The circuit shown below is the



Spar
POLYTECHNIC

Select one

- ☐ a. Parity checker
- ☐ b. None of these
- ☐ c. Full addtractor

Question 28

Incorrect

Mark 0.00 out of 2.00

Flag question

Which of the following gates, are added to the inputs of the OR gate to convert it to the NAND gate?

Select one:

- ☒ a. OR ~~X~~
- ☐ b. AND
- ☐ c. XOR
- ☐ d. NOT

The correct answer is: NOT

Question 29

Incorrect

Mark 0.00 out of 1.00

Flag question

A combinational logic circuit which generates a particular binary word or number is

Select one:

- ☐ a. Decoder
- ☒ b. Multiplexer ~~X~~
- ☐ c. Demultiplexer
- ☐ d. Encoder

The correct answer is: Decoder

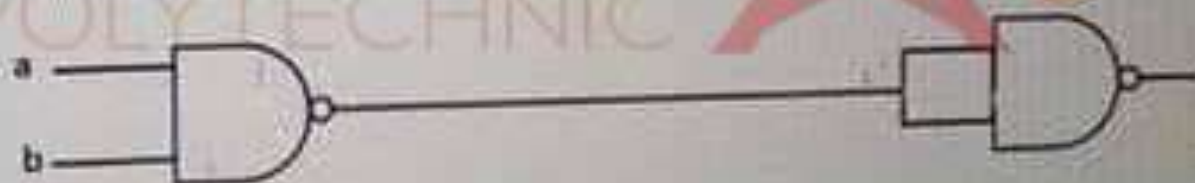
Question 30

Incorrect

Mark 0.00 out of 2.00

Flag question

The combinational circuit given below is implemented with two NAND gates. To which of the following individual gates is it equivalent?



Select one:

- ☒ a. OR ~~X~~
- ☐ b. XOR
- ☐ c. NOT
- ☐ d. AND

Your answer is incorrect.

Question 4

Answered

Mark 5.00 out of 5.00

Flag question

A stable multivibrator is used as

Select one

- ☐ a. Demultiplexer
- ☐ b. Frequency to voltage converter
- ☐ c. Comparative circuit
- ☒ d. Voltage to frequency converter X

The correct answer is: Comparative circuit

Question 5

Answered

Mark 5.00 out of 5.00

Flag question

To implement the Boolean function $F = A \oplus B \oplus C$ we can use a 3 to 8 decoder and connect

Select one

- ☐ a. 5 different lines from its output to an OR gate
- ☒ b. 3 different lines from its output to an OR gate X
- ☐ c. 4 different lines from its output to an OR gate
- ☐ d. 6 different lines from its output to an OR gate

The correct answer is: 4 different lines from its output to an OR gate

Question 6

Answered

Mark 5.00 out of 5.00

Flag question

How many normal functions of basic logic gates (NOT AND OR) need to implement the Boolean Expression?

Select one

- ☐ a. One NOT, one OR
- ☐ b. One NOT, one AND
- ☒ c. Two NOT, one OR X
- ☐ d. Two NOTs, one OR, one AND

The correct answer is: One NOT, one OR

Question 7

Answered

Mark 5.00 out of 5.00

Flag question

Which of the following expressions is not equivalent to $A \oplus B$?

Select one

- ☒ a. $A \oplus B \oplus 1$ X

- ☐ b. AND
- ☐ c. XOR
- ☒ d. NOT ✓

The correct answer is: NOT

Question 2

Incorrect

Mark 0.00 out of 1.50

Flag question

In a positive edge triggered JK flip-flop, a low J and low K produces

Select one:

- ☐ a. No change
- ☐ b. None of these
- ☒ c. High state ✗
- ☐ d. Low state

The correct answer is: No change

- ☐ b. with a counter
- ☐ c. with a multimeter
- ☐ d. with a speaker

What additional feature have a sequential circuit, compared to a combinational circuit?

Select one:

- ☐ a. a memory state
- ☐ b. a power transistor
- ☐ c. AND or OR gates
- ☐ d. clock pulse generators

What is the function of a decimal-binary encoder?

Select one:

- ☐ a. to convert from binary to decimal code

- ☒ a. a memory state ✓
- ☐ b. AND or OR gates
- ☐ c. clock pulse generators
- ☐ d. a power transistor

The correct answer is: a memory state

Which logic gates can be used to make a flip-flop?

Select one:

- a. OR
- b. 1 NAND
- c. EXCLUSIVE OR
- ☒ d. AND ✗

The correct answer is: 1 NAND

Which kind of code must be supplied to the inputs of a BCD-to-

An RS flip-flop is the basic type of sequential logic circuits.

How many flip flops we have?

Answer: 4 flip flops



What is the function of a binary-decimal coder?

Select one:

- a. to input data in binary way into a Analog system
- b. to convert a binary number into decimal one
- c. to count
- d. to input data in binary way into a digital system

Which logic gates can be used to make a flip-flop?

Select one:

- ☐ a. OR
- ☐ b. AND
- ☐ c. EXCLUSIVE OR
- ☐ d. 1 NAND



How many flip flops we have?

Answer: 4 flip flops

The correct answer is 4



What is the function of a binary-decimal coder?

Select one:

- a. to input data in binary way into a Analog system
- b. to convert a binary number into decimal one ✓
- c. to count

✓ View Answer

2. There are _____ cells in a 4-variable K-map.

- a) 12
- b) 16
- c) 18
- d) 8

^ View Answer

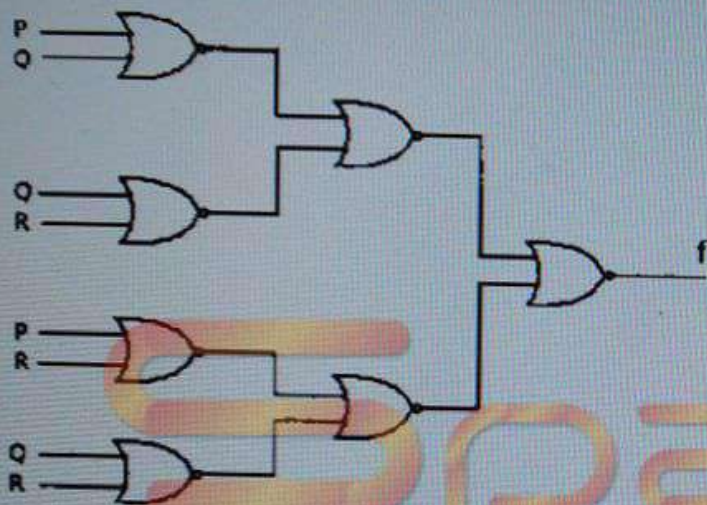
Answer: b

Explanation: There are $16 = (2^4)$ cells in a 4-variable K-map.

3. The K-map based Boolean reduction is based on the following



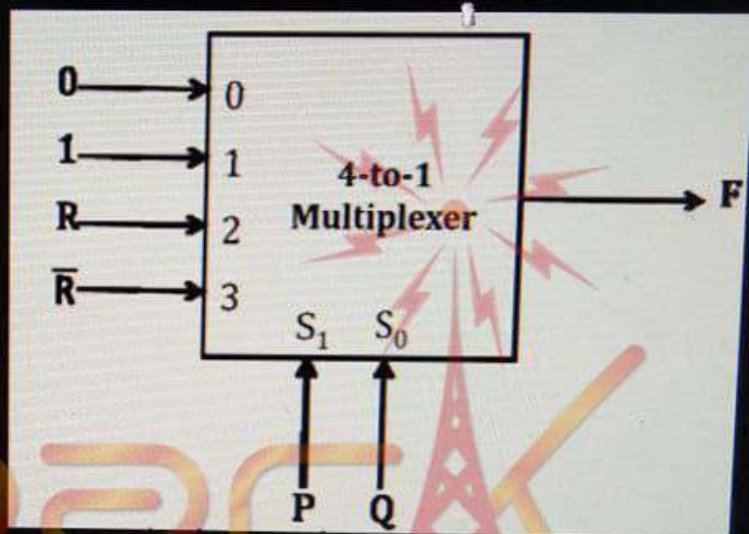
What is the Boolean expression for the output f of the combinational logic circuit of NOR gates given below?



- (A) $\overline{Q + R}$ (B) $\overline{P + Q}$ (C) $\overline{P + R}$ (D) $\overline{P + Q + R}$

Answer: (A) $\overline{Q + R}$

Consider a 4-to-1 multiplexer with two select lines S_1 and S_0 , given below

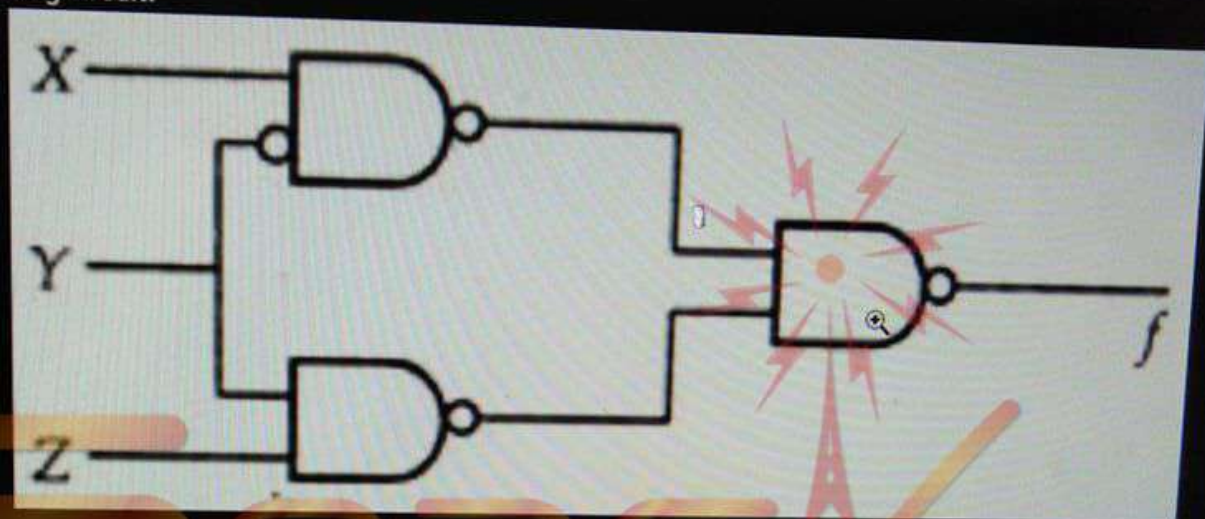


The minimal sum-of-products form of the Boolean expression for the output F of the multiplexer is

- (A) $P'Q + QR' + PQ'R$
- (B) $P'Q + P'QR' + PQR' + PQ'R$
- (C) $P'QR + P'QR' + QR' + PQ'R$
- (D) PQR'

Answer: (A)

Consider the following circuit.



Which one of the following is TRUE?

- (A) f is independent of X
- (B) f is independent of Y
- (C) f is independent of Z
- (D) None of X, Y, Z is redundant

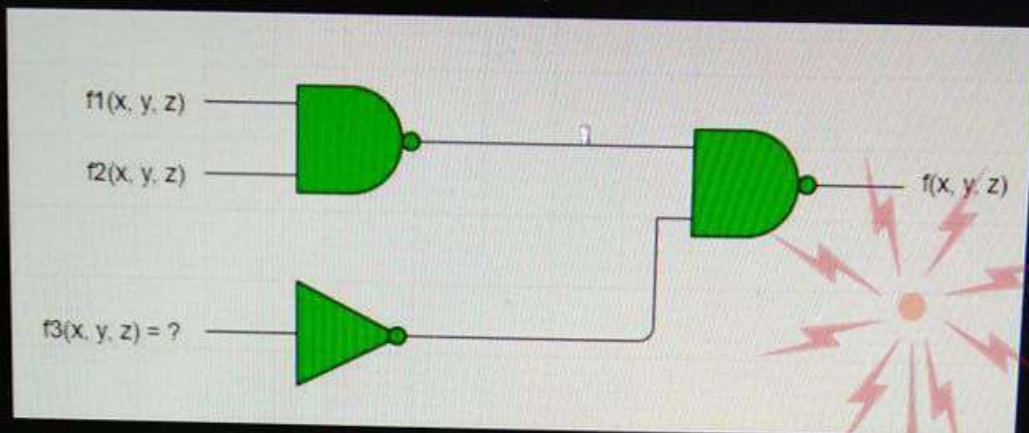
Answer: (D)

Explanation: $[(XY')' \text{ NAND } (YZ)'] = [(XY')' \cdot (YZ)']'$

$$= XY' + YZ$$

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Consider the following logic circuit whose inputs and function and output is f.



Given that

$$f_1(x, y, z) = \sum(0, 1, 3, 5),$$

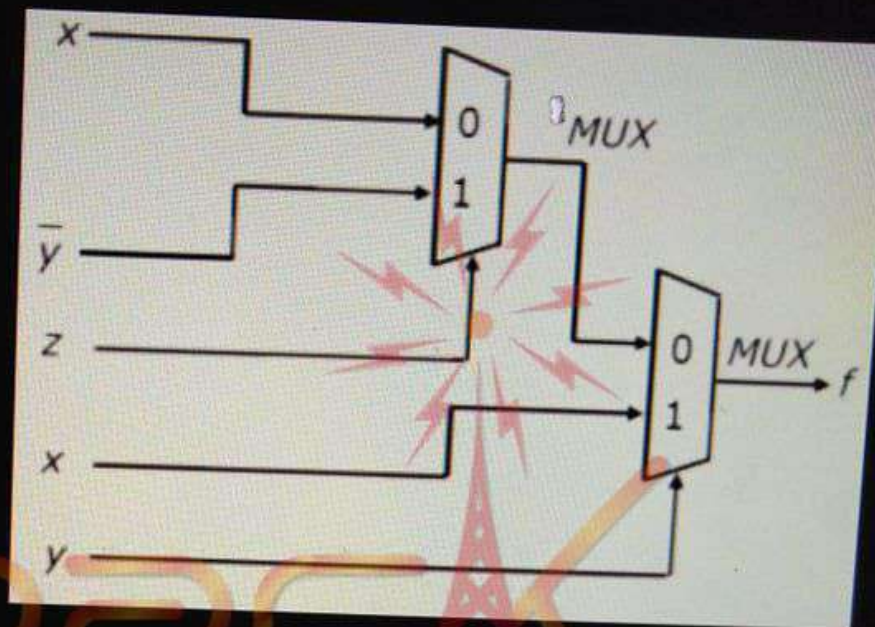
$$f_2(x, y, z) = \sum(6, 7) \text{ and}$$

$$f(x, y, z) = \sum(1, 4, 5),$$

f_3 is:

- (A) $\sum(1, 4, 5)$
- (B) $\sum(6, 7)$
- (C) $\sum(0, 1, 3, 5)$
- (D) None of these

Answer: (A)



Consider the circuit above. Which one of the following options correctly represents $f(x, y, z)$?

- (A) $xz' + xy + y'z$
- (B) $xz' + xy + (yz)'$
- (C) $xz + xy + (yz)'$
- (D) $xz + xy' + y'z$

Answer: (A)

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Explanation: Output from MUX 1 is $x\bar{y}$.

GATE CSE 1999 | Question: 1.8

19

Which of the following functions implements the Karnaugh map shown below?

2.9k V

AB \ CD	CD			
	00	01	11	10
00	0	0	1	0
01	X	X	1	X
11	0	1	1	0
10	0	1	1	0

A. $\bar{A}B + CD$

B. $D(C + A)$

C. $AD + \bar{A}B$

D. $(C + D)(\bar{C} + D) + (A + B)$

gate1999

digital-logic

k-map

easy

GATE | GATE-CS-2014-(Set-3) | Question 65

Last Updated : 25 Nov, 2020

Consider the following minterm expression for F :

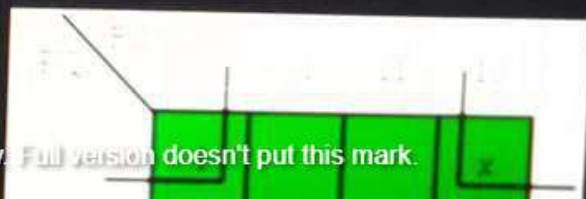
$$F(P,Q,R,S) = \sum (0, 2, 5, 7, 8, 10, 13, 15)$$

The minterms 2, 7, 8 and 13 are 'do not care' terms. The minimal sum-of-products form for F is :

- (A) $QS' + Q'S$
- (B) $Q'S' + QS$
- (C) $Q'R'S' + Q'RS' + QR'S + QRS$
- (D) $P'Q'S' + P'QS + PQS + PQS'$

Answer: (B)

Explanation:



The simplified SOP (Sum Of Product) form of the boolean expression $(P + Q' + R') \cdot (P + Q' + R) \cdot (P + Q + R')$ is

- (A) $(P' \cdot Q + R')$
- (B) $(P + Q' \cdot R')$
- (C) $(P' \cdot Q + R)$
- (D) $(P \cdot Q + R)$

Answer: (B)

Explanation: See following :

$$(P + Q' + R') \cdot (P + Q' + R) \cdot (P + Q + R') = \sum (3, 2, 4) = \sum (0, 4, 5, 6, 7)$$

	$(QR)'$	$Q'R$	QR	QR'
P'	0 1	1	2	3
P	4 1	5 1	7 1	6 1

In the following truth table, $V = 1$ if and only if the input is valid.

Inputs				Outputs		
D_0	D_1	D_2	D_3	X_0	X_1	V
0	0	0	0	x	x	0
1	0	0	0	0	0	1
x	1	0	0	0	1	1
x	x	1	0	1	0	1
x	x	x	1	1	1	1

What function does the truth table represent?

- (A) Priority encoder
- (B) Decoder
- (C) Multiplexer
- (D) Demultiplexer

Answer: (A)

Explanation: Since there are more than one outputs and number of outputs is less than inputs, it is a Priority

$V=1$ when input is valid and for priority encoder it checks first high bit and ignored. Except all are having at 1 represents the "don't care" as we have found a high bit already. So answer is (A).

View Answer

7. Comparators are used in _____

- a) Memory
- b) CPU
- c) Motherboard
- d) Hard drive

View Answer

Answer: b

Explanation: Comparators are used in the central processing unit (CPU) part of the computer. The ALU (Arithmetic Logic Unit) is the part of the CPU that performs arithmetic and logical operations. The ALU uses comparators to compare the results of its operations with the values stored in the registers.

DIGITAL LOGIC DESIGN LAB-7

DIGITAL LOGIC DESIGN LAB-7 ► General ► final

Question 2

Not yet answered

Marked out of 2.00

Flag question

fill the state table

JKQQ+
00 0 ????

Select one:

- ☐ a. Q
- ☒ b. 1
- ☐ c. 0
- ☐ d. Q'



What is the minimal form of the Karnaugh map shown below? Assume that X denotes a don't care term.

cd \ ab	00	01	11	10
00	1	X	X	1
01	X			1
11				
10	1			X

- (A) $b'd'$
 (B) $b'd' + b'c'$
 (C) $b'd' + a'b'c'd'$
 (D) $b'd' + b'c' + c'd'$

Answer: (B)

Explanation: Protected with free version of Watermarkly. Full version doesn't put this mark.